Master Thesis

Sentiment Analysis on Generative Large Language Models based on Social Media Commentary of Industry Participants

MSc Business Administration and E-Business

Supervisor:

Abid Hussain

Submission Date:

15.05.2023

Number of standard pages:

83

Number of characters:

164 654

Authors:

Oana-Alexandra Miron	127338	oami18ab@student.cbs.dk
Khant Nyein Htet Wai	155480	khwa22ab@student.cbs.dk

Glossary

The following glossary represents a list of important abbreviations, ideas, and terms that the readers continuously encounter in this work:

AI	Artificial Intelligence
CSS	Computational Social Science
DOI	Diffusion of Innovation
GPT	Generative Pre-Trained Transformer
GLLM	Generative Large-Language Model
ML	Machine Learning
SML	Supervised Machine Learning
UML	Unsupervised Machine Learning
NLP	Natural Language Processing
SA	Sentiment Analysis
TSA	Twitter Sentiment Analysis

Abstract

OpenAI released their new product, ChatGPT on the 30th of November 2022. ChatGPT is a generative large language AI model and an extension of GPT-3.5, that makes the interaction between the user and the machine more accessible to the general public. The AI-powered chatbot responds to user requests with almost real-time dialogue in a naturally spoken way. The use cases of ChatGPT for businesses are long discussed across media and within business environments. These are manifold and include creating source code, producing creative texts, and assisting in repetitive tasks. With such promising prospects, ChatGPT managed to reach a 100 million user base in record time. Thus, the purpose of this paper is to explore which industries are more open to accepting the technology of generative large language models based on opinions about ChatGPT. Since Twitter became a public forum where well-known figures and regular people are expressing their opinions alike, this paper conducts topic modeling and sentiment analysis on tweets related to ChatGPT. The aim is to map the authors of these tweets to industries and to subsequently uncover the prospects of acceptance of the technology within these different industries. The research finds that the people expressing their opinion about ChatGPT on Twitter either do not relate their profiles to an industry or are part of industries such as: Education & Academia, Entrepreneurship & Business Leadership, Finance & Crypto, and more. While tweets vary from industry to industry and from user to user, this research showcases that the general sentiment over generative large language models is a positive one, which is then discussed from the perspective of technology acceptance and its implications. This paper contributes to technology acceptance research, exhibiting the use of Twitter data to explore perception and acceptance in relation to generative large language models.

Table of Contents

Glossary 2
Abstract 3
Table of Contents 4
Table of Figures 5
Table of Tables 6
1. Introduction 8
1.1. Research Context 8
1.2. Motivation 10
1.3. Research Question 13
1.4. Outline 14
1.5. Conceptual Framework 15
2. Literature Review 17
2.1. Search Mechanism 17
2.2. Generative Large Language Models 19
2.3. Perceptions over Generative Large Language Models 22
2.4. NLP Techniques to analyze Social Media Commentary 27
2.5. Concluding Remarks on Literature Review 34
3. Theoretical Framework 35
3.1. Fundamentals in Diffusion and Acceptance of Technology 35
3.2. Diffusion of Innovation Theory 37
4. Methodology 42
4.1. Research Philosophy 43
4.2. Research Approach 44
4.3. Research Strategy 44
4.3.1. Computational Social Science 44
4.3.2. Netnography 45
4.4. Methodological Choice 46
4.5. General Assumptions and Considerations 47
4.6. Data Collection and Processing 49
4.6.1. Data Collection & Anonymization 52
4.6.2. Pre-Processing for Topic Modeling 54
4.6.3. Topic Modeling 57
4.6.4. Sentiment Analysis on Tweets 60
4.7. Exploratory Data Analysis 62

4.8. Summary of Methodology 63
4.9. Quality Evaluation 64
4.9.1. Validity 64
4.9.2. Reliability 65
5. Findings 66
6. Discussion 77
6.1. Summary of Main Findings 77
6.2. Implications 81
6.3. Strengths and Limitations 83
6.3.1. Strengths 83
6.3.2. Limitations 84
7. Conclusion 86
7.1. Future Research 88
8. References 90

Table of Figures

Figure 1 - Sentiment Analysis Generic Procedure adapted from Wankhade et al. (2022) 28
Figure 2 - Theory of Reasoned Action adapted after Fishbein & Ajzen (1975) 36
Figure 3 - The Five Categories of Adopters and the Adoption Curve (Rogers, 2003) adapted by Wikipedia (2013) 38
Figure 4 - Innovation Decision-making Process, based on Diffusion of Innovations (Rogers, 2003) 41
Figure 5 - Research Onion adapted after Saunders (2009) 43
Figure 6 - Research Methodology Process adapted from Radford & Joseph (2020) 46
Figure 7 - LDA Coherence Score by Number of Topics in steps of 2 59
Figure 8 - Summary of Methods 64
Figure 9 - Top 10 Words of LDA Topic Modeling on User Descriptions by Category 67
Figure 10 - Distribution of Twitter Users by Industry 68
Figure 11 - Number of Users by Sentiment 69
Figure 12 - User Sentiment by Industry 70
Figure 13 - Average Compound Score vs. Tweets Count 72
Figure 14 - Average Compound Score vs. Tweets Count by Industry 73
Figure 15 - Average Compound Score by Industry Over Time 74
Figure 16 - Top 10 Words of LDA Topic Modeling on Tweets by Category 75

Table of Tables

Table 1 - Search Queries by Topic of Interest 18
Table 2 - Accuracy Measures 31
Table 3 - Technical Process of Data Collection and Analysis 50
Table 4 - User-specific Data 56
Table 5 - Tweet-specific Data 56
Table 6 - VADER Accuracy Scores on ChatGPT Tweets Dataset 61
Table 7 - Sentiment Polarity Statistics by Industry 71
Table 8 - Polarity Statistics for Negative and Positive Perceptions by Industry 71

Acknowledgments

We would like to express our deepest gratitude to our thesis supervisor, Abid Hussain (Ph.D) for his mentorship, support and invaluable advice throughout the thesis period. His constructive feedback and expertise have been instrumental in shaping this research.

Words can also not express how much our families and friends supported us with their love, encouragement and understanding during this challenging and rewarding experience.

1. Introduction

The first chapter formulates the research context, the authors' motivation as well as the overall outline of this project. Additionally, a short overview of terms and concepts are provided to introduce concepts closely related to the present research.

1.1. Research Context

With tremendous investments and developments made in the field of artificial intelligence (AI) since the 1940s, the interest in understanding this technology has grown. Consequently, it became a topic of research and discussion due to its potential to shape the future world of business and the way we work. Advancements in Natural Language Processing (NLP) techniques, a subfield of AI, have led to the invention of chatbots and have improved communication between humans and machines through language (Mohamad Suhaili et al., 2021). The applications of chatbots aim at understanding written and spoken language, for instance, to support humans in task automation, information extraction, user guidance etc.

The first pioneering chatbot worth mentioning was developed by the Massachusetts Institute of Technology (MIT) called ELIZA in 1966 (Weizenbaum,1996). Naturally, ELIZA had weaknesses around limited contextual understanding and personalization. Exponential development in the last 10 years in NLP and machine learning (ML) have led to a ground-breaking point where large language models can converse, learn and respond in a humanistic manner and generate respectable outputs (Law, 2022).

A prominent way to measure a machine's intelligence is the Turing Test, which was introduced in 1950 by the British mathematician Alan Turing. For this test, a human interacts with a computer in text-form. If the responses from the computer are indistinguishable to another human and prove capabilities of thinking like a human being, the machine would be said to have passed the test. Two AI models that would make great contenders for passing the Turing test are Google's Bard and OpenAI's ChatGPT (Trincieri, 2023), which have made massive waves inside and outside of the tech industry, across the globe, and caused a re-emergence of hype around AI.

Both ChatGPT and Bard fall into the sub-category of NLP called generative pre-trained language models. Generative pre-trained models are neural networks trained on large amounts of data and fine-tuned on smaller samples of data. These models are a result of the Transformer training architecture, proposed by Radford et al. (2018) in the paper "Improving Language Understanding by Generative Pre-Training".

According to Statista (2023a), the rise in interest in "generative AI" on Google Search happened simultaneously with the launch of ChatGPT, proving that OpenAI's product transformed AI, and especially generative models, into mainstream topics. This is why ChatGPT will be used as a proxy for generative large language models (GLLM) in the present research paper. GPT stands for "Generative Pre-Trained Transformer", however one opinion piece claims it could stand for "General Purpose Technology", based on the way this innovation has the potential to enable productivity boost across industries and occupations, similar to how steam engines, electricity, and computing have done it (The Economist, 2023). According to a recent study by Acumen Research and Consulting (2022), the global generative AI industry will be worth $110.8 billion by 2030.

OpenAI's ChatGPT was released on the 30th of November 2022 and is open for public usage. According to Neumann et al. (2023) the chatbot gives a glimpse on what is expected to come with AI in the near future. While some people share their fascination, others are concerned about how almost omniscient human-like (even though not always factually correct) AI became and how it will affect our lives in the future (Shankland, 2023). People are in awe about how ChatGPT is able to explain any concept to any level defined by the user, write full functional code with a few prompts and diagnose and propose the right treatment. Others worry that this powerful AI can be misused for propaganda, cybercrime, and other harmful purposes (Europol, 2023).

Regardless of the controversies sparked by ChatGPT, GLLMs seem to be on the way to changing how businesses operate. The integration of OpenAI's large language model into Microsoft's search engine, Bing (Mehdi, 2023), is just one of the examples underlining that generative models are changing businesses across the board. Another example, complementary to the controversial opinions related to ChatGPT and education, is the integration of ChatGPT and Duolingo, a company that serves 50 million monthly users as

a language learning platform (OpenAI, 2023a). This case shows that, even though generative models pose a new challenge to traditional education means such as universities (Wood, 2023), novel educational platforms such as Duolingo seem to welcome the technology as an opportunity.

Exploring the acceptance of an emergent technology can provide insights on its potential success or failure, anticipating whether this new technology will persist, or become obsolete in the near future. Additionally, understanding the acceptance will generate insights into which industries are innovation leaders and which are lagging in adoption. The latter is significant to uncover so that decision makers sitting in those industries do not miss out on new opportunities, fail to capitalize on potential revenue streams and meet new customer expectations.

In this context, the present paper wants to uncover to which industries people belong that show interest in GLLMs, and whether their opinions are positive, negative or neutral towards the technology. As a result, the potential for acceptance of the technology can be analyzed based on the identified industries and the determined sentiments.

1.2. Motivation

The motivation behind this study is grounded in numerous sources such as the existing literature on GLLMs, and the current interest in the technology across media, businesses and more.

Firstly, in conducting the systematic literature review, it was found that there are many different viewpoints covered in academic papers on GLLMs. Many studies, including Vaswani et al. (2017) and Brown et al. (2020), have concentrated on the technological advancements in the field of natural language processing that have led to the creation of state-of-the-art models. These models are the backbone of the latest chatbots, such as ChatGPT, due to their capacity to communicate with users in versatile human language. This level of versatility and interaction between user and GLLMs could be understood as one of the factors that triggered extensive studies on the models' explainability. The algorithms underlying GLLMs can be categorized as "black boxes" according to Danilevsky

et al. (2020) because they are primarily built on deep learning techniques. Consequently, papers like Hovy & Prabhumoye (2021) and Danilevsky et al. (2020) conclude that GLLMs have limited explainability, leaving users unsure of why the models respond in the way they do. Nonetheless, their functionality cannot be disputed, and the limitations of application are relatively unexplored, which is why, for example, ChatGPT is becoming the subject of countless research studies from different fields. Over 1000 studies have been published since 2023 showing that the topic has generated a vast amount of interest among researchers. It was found that very few papers have focused on the user's perceptions of the technology.[1] Since the technology can have broad applications, user perception could become an indicator of willingness to adopt the technology.

The only paper which aims to uncover user perception as of February 2023 is Haque et al. (2022). Haque et al. (2022) employ NLP techniques such as topic modeling and sentiment analysis on tweets to understand the characteristics of early adopters of ChatGPT. Moreover, the paper aims to spot the main topics discussed concerning ChatGPT and what sentiment polarities exist towards ChatGPT intending to showcase which professions are most interested in the chatbot's capabilities (Haque et al., 2022). Two other research papers have also opted to analyze social media content about ChatGPT however, with a different goal. While Taecharungroj (2023) is researching what topics are most commonly discussed in relation to ChatGPT, Balcioglu & Çerasi (2023) use YouTube commentary to demonstrate different options for sentiment analysis on the topic.

Furthermore, the choice of analyzing social media commentary established several compelling matters. Unlike the times of the arrival of other groundbreaking technologies like the first computer or the first email, we now live in an era where an integral part of daily internet usage for over half the earth's population is sharing opinions and participating in public discussions on social media platforms (Statista, 2023b). The peer-to-peer impact is one of the most significant influences on user behavior, according to social media network literature (Bickart & Schindler, 2001). According to Grover et al. (2019), social media platform data, e.g. tweets, may be crucial for understanding user

[1] Results for "ChatGPT" since 2023 on Google Scholar shows 1,290 papers as of March 9th 2023

behavior, and their acceptance of a specific technology, and can have an impact on the acceptance levels. While Twitter is not the biggest social media platform compared to Facebook, Instagram, and TikTok, it is where influential people and organizations break news, and where people share their reactions directly (O'Malley, 2022). It has unmatched cultural power and what has been shared often goes beyond Twitter. Tweets make their way into newspapers and websites and even have the capacity to move stock prices in real-time (Vazquez, 2018).

Thus, social media commentary can become an insightful source to analyze and understand different perspectives and attitudes towards the new wave of changes in people's lives. Identifying early adopter sentiments is crucial for a few reasons. First off, early adopters are devoted and powerful users of a product, and their views and feelings can influence how the general public perceives new technology (Haque et al., 2022). This data can offer crucial insights into the product's likelihood of success or loss. Secondly, because early adopters frequently experience issues or problems with new technology first, their input can be used to find and address these issues before they spread widely (Haque et al., 2022). Researchers are increasingly using Twitter data to obtain previously untapped information, for instance, tweets have been analyzed to learn how the public feeling and willingness to accept new technologies such as the "Internet of Things" (Bian et al., 2016), blockchain (Mnif et al., 2021) and how self-driving cars may disrupt individual mobility (Kohl et al., 2017).

In this context, papers like Haque et al. (2022) prove to be of particular interest, as they use the commentary of users to dive into what characterizes early adopters and their opinions. Haque et al. (2022) has been published at the end of 2022, which implies that the tweets available regarding ChatGPT were limited. Thus, we consider that conducting the present research on a larger data sample can provide a better understanding of the sentiment over GLLMs. Additionally, the present paper assumes perception of different industries participants can be uncovered based on the combining the user descriptions and tweets, thereby shifting from the methodology used in Haque et al. (2022).

Lastly, the extensive interest in the application and consumption of AI in our everyday lives has also reached businesses across the board. This can be cemented by prompt

reactions from the largest technology firms like Google and Microsoft, which devoted billions of dollars to further development and implementation of their AI models since the buzz around ChatGPT (Forbes, 2023). Albeit there is a flood of news and updates from different industries' leading figures and progress around artificial intelligence daily, generative AI took the spotlight since the last month of 2022. Additionally, only five days after the release of ChatGPT, a million daily users have been reached, causing the service to be unavailable at peak times. According to a report note from UBS (Union Bank of Switzerland), the actively unique user visits for ChatGPT has reached 100 million just two months after its launch, making it the fastest-growing consumer application to date and placing it at the forefront of the upcoming shift (Hu, 2023). Thus, it can be concluded that ChatGPT has put AI and GLLMs into the spotlight, causing tremendous interest among researchers, the media and users across the world.

Thereby, the present paper hopes to build on the existing literature related to generative large language models by analyzing social media commentary to gain insights on the sentiment of different industries participants over the technology. Based on the popularity presented in research papers, the media and users' opinions, the present paper wants to provide insights on the sentiment polarity of users from different industries as an indicator for the willingness of technology adoption.

1.3. Research Question

Based on the provided motivation, this paper strives to provide an answer to the following research question:

How are generative large language models perceived by participants of different industries based on social media commentary?

To support with structuring the process of answering to the main research topic, the following sub-research questions (Sub-RQ) have been developed:

Sub-RQ1: To which industries are users that take part in the social media commentary about ChatGPT belonging?

The objective of Sub-RQ1 is to uncover potential industries that are particularly keen to express their opinion on ChatGPT. Once the industries have been identified through Sub-RQ1, the next question that leads to Sub-RQ2 is which sentiment polarities (e.g. Positive, Negative, or Neutral) each sector holds towards ChatGPT.

Sub-RQ2: How is ChatGPT perceived across the identified industries?

The goal of Sub-RQ2 is to obtain the ratio of favorable, neutral, and negative opinions on ChatGPT. Another reason for exploring the sentiment is that perception can be an indicator of adoption, which is why Sub-RQ3 deals with analyzing which industries are adopting this new technology the most.

Sub-RQ3: What do tweets about ChatGPT indicate about the acceptance of GLLMs?

The objective of Sub-RQ3 is to scrutinize the prior results from a diffusion theory perspective and sociological point of view, to anticipate impacts on business and society.

1.4. Outline

This research paper is divided into seven main sections: Introduction, Literature Review, Theoretical Framework, Methodology, Findings and Analysis, Discussion, and Conclusion. A depiction of the research context, motivation, and the research questions standing at the core of the present research is presented in the introductory chapter. Additionally, the last section of the introductory chapter provides the user with explanations of the key technical concepts related to this paper.

The second section of the paper details the literature review conducted in relation to the present study. The search mechanisms employed, the current knowledge of the studied topics, and its enrichment opportunities are described in this section. Following is the theoretical framework. The third section provides the reader with the theories and concepts that form the theoretical backbone of this research. Once there is a consensus about the theoretical and empirical background, the research methodology is introduced. The methodology section details the philosophical drawings of the research, their implications, the methods employed in data collection and analysis together with the

limitations of the present research. The fourth section details the findings resulting from analyzing the data. This section serves as the foundation for answering the sub-research questions and the main research question. The discussion part then explains the findings in relation to the research questions, concluding with answers to these. Additionally, further interpretations of the findings through the lenses of the theoretical background and the existing literature are provided. Finally, a conclusion for the study is stated together with proposals for further research.

1.5. Conceptual Framework

The present section introduces concepts closely related to the present research and often referred to, throughout the paper.

Artificial Intelligence (AI)

Artificial Intelligence is a subfield of Computer Science that aims to make machines intelligent, enabling them to function appropriately with foresight in their environment (Nilsson, 2009). This implies making machines capable of learning, solving problems, making decisions, and executing tasks that usually require human cognition.

Machine Learning (ML)

Machine learning, a sub-branch of AI, that focuses on creating systems that can learn from data without specific programming. Machine learning algorithms make predictions or decisions based on the incoming data by using statistical techniques to find patterns (Jordan & Mitchell, 2015).

In supervised machine learning (SML), the algorithm is trained on labelled data, which means that the incoming data has already been connected to the desired outcome (Kühl et al., 2022).

In unsupervised machine learning (UML) the methods and algorithms are trained on data that has not been labelled, indicating that the input data has no previously known relationship to the output. Unsupervised learning aims to reveal structures and patterns

in the raw data without knowing the final product and does not necessarily have a correct solution (Kühl et al., 2022).

Natural Language Processing (NLP)

Natural Language Processing is a branch of AI that deals with the natural interaction between humans and machines through human language. NLP aims to make it possible for computers to process, interpret, and produce human language in text or spoken words, enabling them to carry out tasks like sentiment analysis, question-answer interactions, and language translation (Jurafsky & Martin, 2000).

Generative Large Language Models (GLLMs)

Generative Large Language Models are models trained to generate content for specific NLP tasks such as translation, question-answering, etc. (Shen et al., 2023; Vaswani et al., 2017).

In the context of this paper, ChatGPT will be considered a proxy for GLLMs.

Topic modeling

Topic modeling is an unsupervised machine-learning technique that finds hidden semantic structures in a big corpus of text. The objective of topic modeling is to recognize the fundamental themes that are covered in the corpus and to determine relationships between themes. The frequency of words and phrases in the text is examined using statistical techniques by topic modeling algorithms, which then classify text into themes based on their patterns of co-occurrence (Tong & Zhang, 2016).

Sentiment polarity classification

In the context of this paper, the terms “sentiment polarity classification” and “sentiment analysis” are used interchangeably. For the scope of this paper, these are defined as the classification of sentiments based on their orientation. The three orientation categories are negative, positive, and neutral (Wankhade et al., 2022).

Sentiment compound score

The sentiment polarity compound is a metric used in sentiment analysis that represents the overall emotional tone of a text, in this case a tweet. The compound polarity can be thought of as an aggregate measure of all the other sentiments, normalized to be between −1 and 1. A compound score of 1 indicates extremely positive sentiment, -1 indicates extremely negative sentiment (Pano & Kashef, 2020).

Web Scraping

The practice of automatically gathering data from web pages is known as web scraping. Scrapers, also known as crawlers, are computer programs employed in this task and have the ability to extract large amounts of data from web pages and social media platforms (Mitchell, 2018).

2. Literature Review

The present chapter details the literature review conducted as part of the thesis. The chapter includes a description of the search strategy chosen, a review of the relevant papers, and concluding remarks based on the academic literature reviewed.

2.1. Search Mechanism

To conduct the literature review, three research areas were identified and prioritized:

1. Generative Large Language Models
2. Perceptions over GLLMs
3. NLP techniques to analyze social media commentary

Relevant academic articles related to the abovementioned topics were identified based on relevant keyword queries (a non-exhaustive list of keywords can be found in Table 1).

Research Field	Keywords
Generative Large Language Models	*Artificial Intelligence based on NLP, Generative AI, Generative Language Models, Large Language Models, Natural Language AI, ChatBots, Generative ChatBots, ChatGPT, BARD, Transformer Model, BERT, Generative AI Models, Pre-trained models, Pre-trained Generative Models, AI Chatbot, Chatbots Language Model*
Perceptions over Generative Large Language Models	*Opinion on Generative AI, Opinions on Language Models, Large Language Models Analysis, Analysis of Generative AI, Industry perception over ChatGPT, Industry perception over Large Language Models, Implications of Large Language Models, Implications of Generative AI*
NLP techniques to analyze social media commentary	*Topic Modeling, NLP for Social Media, Topic Modeling on Twitter Description, Twitter Commentary Analysis, Sentiment Analysis, Tweets Analysis, Twitter Sentiment Analysis, Social Media Analysis*

Table 1 - Search Queries by Topic of Interest

Based on the identified keywords, initial searches were made on Google Scholar, LibSearch, and Det Kgl. Bibliotek (online service of the Danish National Library). Derived from how much the content of the found papers intersected with the three themes this paper covers, academic articles were then grouped into three categories (high relevance, moderate relevance, and low relevance). Papers that did not fit in any of the categories were disregarded.

As a second step in the literature review, the backward snowballing (Wohlin, 2014) method was employed, whereby the references section of relevant articles was scanned in search of additional relevant research papers. The papers found through the snowballing method have then been run through the same process of identifying their degree of relevance to the topics of this paper.

Additional weight was attributed to papers that were covering ChatGPT in connection to any of the themes discussed in the present paper since ChatGPT has been used as a proxy for GLLMs in the current research. The publication year has also been a factor in weight attribution. It was, therefore, considered that the research fields covered in this paper have developed significantly throughout the years, making certain older publications redundant to the present state of GLLMs, opinions over them, and the field of sentiment analysis. Thus, with few exceptions, the scientific papers dated closer to 2023 were attributed a higher weight. It is worth mentioning, however, that the number of citations has not played a definitory role in the assessment of papers, as some of the papers

covering GLLMs have been published in the past two years and have yet to be cited in high numbers.

The language of publication of the papers considered for this literature review is limited to English.

2.2. Generative Large Language Models

Generative Large Language Models (GLLM) are a subset of Large Language Models (LLM), with the terms being used interchangeably at times. However, a main distinction between the two has to be made: LLMs are trained and used for multiple NLP tasks (text generation, classification, matching, translation, structured prediction etc. (Li, 2018), while GLLMs are task-specific (Shen et al., 2023; Vaswani et al., 2017).

When studying academic literature concerning GLLMs and LLMs the dependency of the topic to its parent fields of NLP and AI has become highly noticeable. With numerous papers being published on NLP developments such as Gehring et al. (2017), Kalchbrenner et al. (2017) and Kim et al. (2017), it has been established through literature that deep learning and the concept of attention have increased the performance of language models across various NLP tasks including abstractive summarization, and comprehension (Vaswani et al., 2017). In this context, one of the major developments stands in the introduction of the Transformer architecture with the “Attention is all you need” publication by Vaswani et al. in 2017. The study proposes the new transformer architecture which relies entirely on self-attention to map its input and output. The proposed architecture moves away from Recurring Neural Networks, that analyze each word in a sequence at a time and introduces the concept of attention which enables the faster development of LLMs. Following to Vaswani et al. (2017), two papers Radford et al. (2018), and Brown et al. (2020) have brought new knowledge and developments that put the grounds for GPT-3, an autoregressive language model with 175 billion parameters, 10 times more than any previous non-sparse language model. Radford et al. (2018), showcase that language models trained to maximize the likelihood of highly varied text corpus begin to learn how to perform a multitude of NLP tasks in an unsupervised

manner, while Brown et al. (2020) introduce GPT-3, the LLM which shows strong performance on many NLP tasks, benchmarking in zero, one or few-shot learning settings. Based on GPT-3, OpenAI has further trained GPT models for various applications such as InstructGPT, and the one used as a proxy for GLLMs in the case of the present paper, ChatGPT (OpenAI, 2022). Other transformer-based models include BERT, developed by Google and further detailed in Devlin et al. (2018).

The technical developments of NLP and AI triggered further research on aspects such as their explainability. While explainability can refer to numerous things, a special focus was placed on the outcome explanation problem, which entails that the user can understand how a model arrives at its results (Guidotti et al., 2018). Danilevsky et al. (2020) present the need to discuss explainability since architectures like transformer (Vaswani et al., 2017) and models like GPT-3 (Brown et al., 2020) or BERT (Devlin et al., 2018) have moved away from white box techniques (e.g. decision trees, linear or logistic regression) to black box techniques such as deep learning and language embedding (Danilevsky et al., 2020). Danilevsky et al. (2020) conclude that there is not enough standardization for explainability to be measured in relation to black-box systems. The paper goes even further to reinforce the importance of explainability in the context of human trust. More specifically, the paper states that the lack of explainability could have consequences such as the loss of user trust in relation to systems such as chatbots or recommendation systems based on deep learning models. A similar perspective is brough by (Hovy & Prabhumoye, 2021).

The literature includes papers which offer examples of GLLMs applicability as a source of inspiration for designers from numerous fields (Yun et al., 2022) or as a tool in areas such as, among others, drug development (Zeng et al., 2022), or couples therapy (Chakravarthula et al., 2015).

Another main applicability is related to chatbots. Papers from Caldarini et al. (2022) and Luo et al. (2021) are both discussing developments in relation to chatbots. These papers provide similar definitions of chatbots which can be understood as conversational agents

that interact with users through natural language by mimicking human behavior. While both papers discuss the advancements brought by transformers and deep learning techniques to chatbots, they both conclude that the technology has not matured yet presenting limitations. Caldarini et al. (2022) state that the main limitation of chatbots is based on their lack of possibility to infer context. More precisely, chatbots are limited to predicting the best fitting output based on an input, without providing reasoning, considering the context, or taking into consideration previous steps of the conversation. Luo et al. (2021) restated the findings of Kocaballi et al. (2020) and Fryer et al. (2017) that identified response propriety and collaboration with humans as the limitations of chatbots. Linking to the developments that led to a new generation of chatbots, Przegalinska et al. (2019) propose a new methodology of chatbot performance measures. Przegalinska et al. (2019) start by acknowledging the already existing performance measures that are often linked to the domain where a chatbot is implemented. Such measurements are for example conversation length, often used to measure chatbots in fintech, or engagement metrics related to purchases, which are often used in relation to retail chatbots. However, the study moves from already existing measures and underlines the importance of trust in chatbots. From this the following new measures are derived: transparency and honesty, integrity, and degree and benevolence. Transparency consists of the ability of the chatbot (agent) to send honest signals in the communication; integrity is associated with credibility and entails that the agent will act consistently with past experiences, thus becoming predictable; and control and benevolence underline the degree to which the intentions of the chatbot are aligned with those of the user (Przegalinska et al., 2019). The authors conclude by underlying the importance of unblack-boxing AI and providing greater explainability to the models on which the chatbots are based (Przegalinska et al., 2019).

An additional research area related to GLLMs and LLMs is due to high developments and their broad application spectrum. GLLMs and LLMs are, thus, extensively covered in academic literature in relation to bias, radicalization, and their relation to legal governance. Hovy & Prabhumoye (2022), found that NLP models can be prone to bias and have identified five sources of bias, in their research: the data, the model, the

annotation process, the input representations and the research design used (e.g. language used in NLP models). McGuffie & Newhouse (2020) conducted their research with a specific focus on GPT-3 and its role in radicalization. Their findings show that GPT-3 has demonstrated to be better than its predecessor model, GPT-2, at generating extremist text in a manner that emulates informational and influential content that has been used in propagating far-right extremist ideologies. OpenAI has taken strong preventive measures to limit the use of GPT-3 in such scopes (Solaiman et al., 2019). However, the paper further goes on to state that there is a risk of a copycat technology, concluding that stakeholders (e.g. policymakers, governments) should investigate how to build social norms, policy, and educational initiatives to avoid machine-generated propaganda and disinformation (McGuffie & Newhouse, 2020). On a similar note, papers from Hacker et. al (2023) and Helberger & Diakopoulos (2023) debate the state of the regulation concerning LLMs and GLLMs. The two papers agree that new regulation is needed to cover AI such as GLLMs and LLMs due to their black-box characteristics.

With numerous studies around the technicalities, implications, and applications of GLLMs it is hard to debate their position as a hot topic in the research community. Another observation that can be made as a result of the systematic literature review is that few studies are focused on the relationship between GLLMs and users, even though the present literature provides an insightful base for research extensions in this realm.

2.3. Perceptions over Generative Large Language Models

GLLMs are becoming more popular due to their impressive capacity to produce output through prompts in text form. The transformer architecture made it feasible to train larger and more potent language models that enable automated content creation. Generative AI is starting to become an industry itself and comprises more revolutionary models than ChatGPT (Gozalo-Brizuela & Garrida Merchán, 2023). Although Gozalo-Brizuela & Garrida Merchán (2023) constitute that the issue of accuracy is way too rudimentary for professionals to rely on this technology, Zarifhonarvar (2023) anticipate substantial short and long-term ramifications of generative AI on the labor market. Zarifhonarvar's study found that 32,8 % of occupations that are mapped in the ISCO

(International Standard Classification of Occupations) could face full impact, which means that there will be a displacement of workers. Naturally, it depends on several factors such as the belonging industry and the availability of reskilling and upskilling possibilities, but workers executing routine jobs are most likely to be affected. This does not mean, that highly skilled professionals might not be affected at all. Guo et al. (2023) were interested in how comparable ChatGPT is to human-experts and their results indicate that in more than half of the questions, particularly in the areas of finance and psychology, ChatGPT's answers are perceived more helpful than those provided by humans-experts. Yet, generative AI has the potential to boost productivity and create new opportunities for high-skilled occupations, too (Zarifhonarvar, 2023). However, Shen et al. (2023) claim that large language models are double-edged swords. While they can quickly tap into data, summarize and provide guidance in diverse use cases, observations show signs of the hallucination effect, which describes confident responses from AI that cannot be justified by its training data. The following sections will demonstrate different perceptions of experts that have investigated the use of ChatGPT in their field, as well as perceptions gathered from the masses via social media.

Content Creation & Media

One heavily discussed field concerns producers of text e.g. professionals working in journalism and media. Pavlik (2023) argues that ChatGPT possesses the ability to pass the Turing test but does not have the ability to be creative in the same way as humans. In the media sector, where AI is widely used in several contexts, Pavlik (2023) states that an AI tool could be a valuable asset for human journalists or media workers to automate the creation of narratives, image recognition, and real-time transcriptions and potentially improve both the quality and efficiency of their work under time constraints. However, with that being said, GLLMs also pose a threat to these workers in times of shrinking budgets and a need for economic efficiency (Pavlik, 2023). Goldstein et al. (2023) stress that GLLMs could dramatically decrease costs for human writers. In conjunction, the study claims that GLLMs might also be misused by malicious actors to spread propaganda and mislead people with high-quality and low-detectability AI-created content (Goldstein et al., 2023).

Software Engineering

In tandem, the software engineering field heavily discusses using generative AI for auto-complete, code translation between computer languages, and the conversion of conversational descriptions into code (Sun et al., 2022). The most well-known example is GitHub's CoPilot, a generative AI co-programming tool. Sun et al. (2022) mentions the possibility that tools for code or co-programming could be a logical fit for conversational agent interfaces or AI assistants. Despite the fact that the output quality of co-programming tools like CoPilot, which initially was based on the GPT-3 model, can be improved, Weisz et al. (2021) claim that software engineers are willing to use such models in their work. When evaluating ChatGPT, Sobania et al. (2023) underline that this new GLLM is outperforming state-of-the-art deep learning approaches in terms of bug fixing.

Furthermore, according to White et al. (2023), ChatGPT has extensive capabilities that are not uncovered fully by software engineers. GLLMs hold great promises for supporting the automation and the speeding-up of routine tasks across the life cycle of software engineering, not only through code creation but also by enabling quick experimentation at numerous levels of abstraction (White et al., 2023). Though at this point, ChatGPT needs to be closely monitored by human users due to its proneness to hallucinate with inaccurate output, prudent prompting can lesson some of these issues. Hence, White et al. (2023) conclude that the ability to integrate prompting skills at various stages of the software life cycle to enhance software engineering must be further researched.

Education & Academia

The new prospects of ChatGPT are also hitting academia and the educational sector. Thorp (2023) raises concerns regarding how ChatGPT will alter education. It can create writings in the style of essays, songs, and even poetry in a matter of seconds. This becomes alarming when you think of the creation of scientific papers, where in recent research, academic reviewers were given abstracts generated by ChatGPT, and only 63% of the fakes were detected (Thorp, 2023). Nonetheless, according to Shen et al. (2023) early reports show that artificial abstracts can be flagged by existing software, but journals in general will have a harder time with scientific integrity.

In response to ChatGPT, there is also an overhaul of curricula and a rethinking of the way teachers and students teach and learn (Huang, 2023). According to Huang (2023), administrators of schools and higher education in the United States are completely redesigning their classes, adding new elements like more oral exams, group projects, and handwritten evaluations in place of typed ones, crafting questions that are too smart to be solved with generative AI. Khalil & Er (2023) have examined 50 academic written essays by ChatGPT and plagiarism-detection software flagged only 10 essays with critical levels of similarity, suggesting that detecting plagiarism needs to shift from a similarity check to verifying the origin of content. Recognizing that ChatGPT has raised concerns, OpenAI has released an AI classifier in January 2023, trying to support educators to distinguish AI-written and human-written text, while stating that it is impossible to detect all AI-written text (OpenAI, 2023b).

In contrast, the capabilities of ChatGPT for helping educators and students are getting discussed just as much as the concerns. Zhai (2022) mentions favorable points around adaptive learning, personalized recommendation, individualized instruction, early identification of learning needs, student record management, grading and assessment and course scheduling. Policymakers around the globe are starting to express their support in integrating ChatGPT and other AI tools into its educational systems, such as the Singaporean government (Leo Wong, 2023) or the Ministry for Schools and Education in North Rhine-Westphalia, that are the first to provide extensive guidelines (Ministry for Schools and Education in North Rhine-Westphalia, 2023).

Exam-taking is another aspect that is polarized. Kung et al. (2023) evaluated the performance of ChatGPT on the United States Medical Licensing Exam (USMLE) and found that it performed at the passing threshold for all three parts of the exam without any specialized training or adaptation. Varanasi (2023) shows further examples of advanced exams that were passed by ChatGPT, including the Stanford Medical School final in clinical reasoning, the MBA exam in Wharton, and four law school courses at the University of Minnesota. While the chatbot received passing grades, it should be noted that it had problems with more complex questions that required an understanding of how multiple inputs and outputs interacted (Varanasi, 2023). Still, Susnjak (2022) undermines that

exam-taking and supervision techniques, especially for online exams need to be re-scrutinized.

General Perception of GLLMs based on Social Media Commentary

At the time of writing the literature review in March 2023, two papers analyzed tweets around ChatGPT to understand what perceptions were formed around it.

Taecharungroj (2023) inspected 233 914 tweets composed in the English language from the 30th of November until the 31st of December using the Latent Dirichlet Allocation (LDA) topic modeling method. Three overall themes were identified: general topics, functional domains, and potential impacts. The themes comprised ten topics, namely: news, technology, reaction, creative writing, essay writing, prompt writing, code writing, answering questions, impacting tech, and impacting humans. The examined tweets revealed that ChatGPT, in general, can have both positive and negative effects on technology and people. Concerns about displacing white-collar professionals and creative workers like software engineers, attorneys, journalists, marketers, creative writers, and storytellers worried a lot of Twitter users. Similar to the previously presented academic studies on the impact of ChatGPT on the education sector, the results show that Twitter users are heavily discussing how to adapt to this rapidly changing environment and best prepare for the future. Regardless of the existing doubts, a significant part of Twitter users cheered the technology's clarity, brevity, helpfulness, speed, and innovativeness, also in comparison to the traditional Google search. Furthermore, Twitter users have shared creative examples of poems and fictional stories produced by ChatGPT, showing that generating creative content based on specific prompts has been perceived as favorable. Moreover, many users think that ChatGPT could significantly enhance business procedures, especially in the area of marketing and legal, where it could simplify the development of content and legal contracts. To conclude, the paper from Taecharungroj (2023) shows that the diffusion of large language models is on the rise in many dimensions and that the results raise more questions than answers.

In the same line of research, Haque et al. (2022) explores 10 732 tweets written in the English language that were collected from the 5th of December 2022 to the 7th of

December 2022. By running LDA, Haque et al. (2022) have identified nine main topics in order to perform a manual in-depth qualitative sentiment analysis of each topic afterwards. The topics identified were "Disruptions for Software Development", "Entertainment and Exercising Creativity", "Natural Language Processing", "Impact on Educational Aspects Chatbot Intelligence", "Impact on Business development", "Implications for Search Engines", "Q&A Testing" and "Future Careers & Opportunities". Using concrete tweet examples for each topic, the study provides a qualitative analysis of the topics addressed in tweets. This was done by analyzing 9 datasets containing 100 randomly chosen tweets for a particular topic. Each tweet in the datasets was assigned a sentiment score of -1, 0, and 1, where -1 stands for negative emotion, 1 for positive sentiment, and 0 for neutral sentiment.

2.4. NLP Techniques to analyze Social Media Commentary

As further elaborated in section 2.2., NLP is a vast field that deals with how computers understand human language. Thus, NLP has multiple sub-branches defined by what they can achieve and what tasks they can handle. Some of the tasks of NLP are language translation, sentiment analysis, text classification and more (Khurana et al., 2023). In this context, the present paper will employ NLP techniques covering the tasks of sentiment analysis and text classification to analyze social media data and answer the research question.

Carvalho & Plastino (2021), Wankhade et al. (2022), and Xu et al. (2022) provide extensive reviews of sentiment analysis (SA). Based on these papers a comprehensive overview of SA can be provided.

Sentiment analysis is used to fulfill the need to analyze text with the goal of inferring sentiment. Wankhade et al. (2022) highlights two main reasons why sentiment analysis is needed by businesses: to analyze customer satisfaction with the products or services the company provides and to better their offering based on customer feedback.

SA, also referred to as opinion mining, is the process of analyzing text or audio to gather people's opinions, sentiments, or impressions with the use of NLP techniques such as

text processing, computational linguistics, text analysis, and more (Carvalho & Plastino, 2021; Khurana et al., 2023; Thelwall et al., 2020; Wankhade et al., 2022).

In conducting a sentiment analysis, each analyzed object is attributed a polarity score (positive, negative and neutral in some cases (Wankhade et al., 2022). Wankhade et al. (2022) explains the levels on which sentiment analysis is conducted: document level, sentence level, phrase level and aspect level. Thus, a single polarity can be assigned on the entirety of a document or on sub-parts of it (sentence, phrase or aspect).

Along with Wankhade et al. (2022), Carvalho & Plastino (2021) provide similar descriptions of the general procedures leading up to sentiment analysis. The following steps are identified: Data Selection, Data Scraping, Data Analysis, Approach and Task.

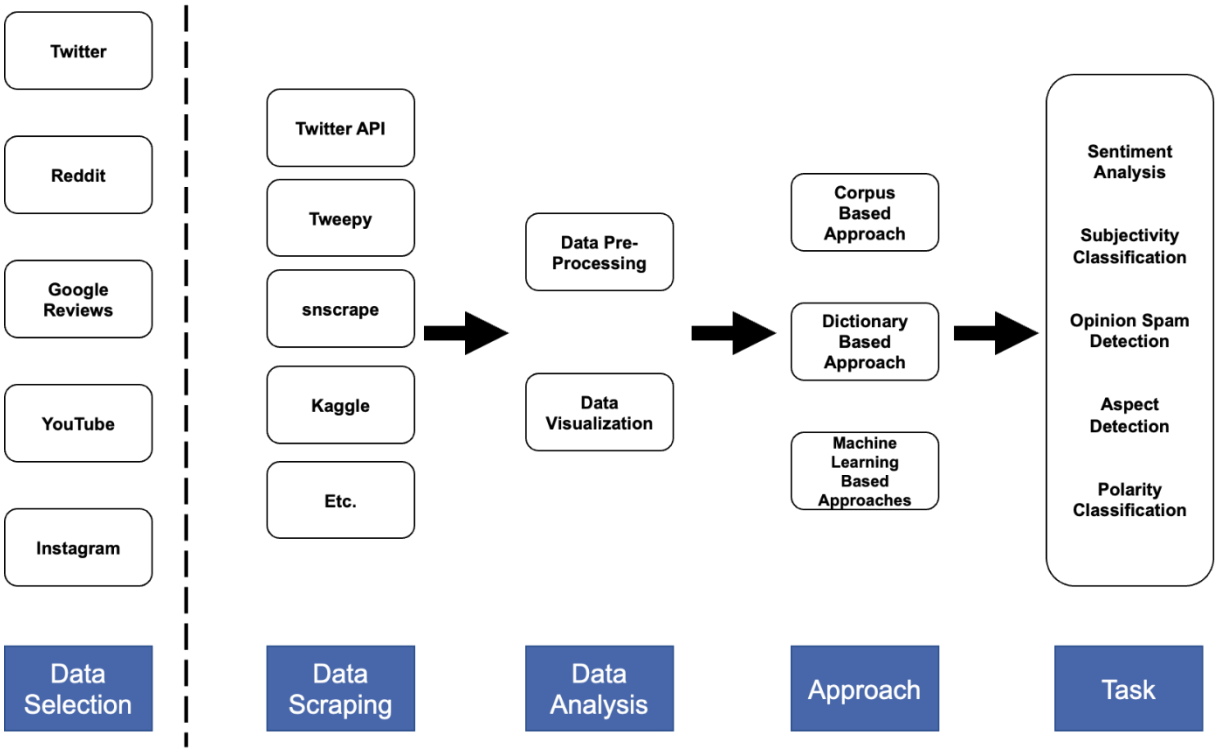

Figure 1 - Sentiment Analysis Generic Procedure adapted from Wankhade et al. (2022)

As part of the SA process, the collection of data from social media is often presented as one of the viable options, with numerous studies stating social media as a dynamic data source from where information about topics of interest can be gathered (Carvalho & Plastino, 2021; Wankhade et al., 2022; Xu et al., 2022). The reasoning for this is that users on platforms such as Twitter, Reddit, and Facebook express their opinions about life

events, products, and services through comments and posts providing the type of data needed for sentiment analysis (Xu et al., 2022).

Once data is collected, SA can be conducted depending on the type of analysis desired. Thus, the tasks of SA cover subjectivity classification, spam detection, polarity detection, cross-domain classification and more (Wankhade et al., 2022). Based on the task of SA, algorithms can be built based on feature extractions and word embedding techniques (Wankhade et al., 2022). The three main categories of algorithms discussed in the literature are lexicon-based, ML-Based, and hybrid approaches (Xu et al., 2022).

The lexicon-based approach involves a collection of tokens where each token is assigned a predefined polarity indicating whether the token is neutral, positive, or negative (e.g., 1: positive; 0: neutral; -1: negative) (Wankhade et al., 2022). Sub-categories of the lexicon-based approaches are corpus-based approaches and dictionary-based methods. If the corpus-based approach employs both semantic and syntactic patterns to ascertain the emotion, the dictionary method consists of a list of predefined opinion words collected manually (Wankhade et al., 2022). The latter method assumes that synonyms have the same polarity as the base word and that antonyms have the exact opposite polarity (Kaity & Balakrishnan, 2020). Lexicon-based techniques are particularly feasible at the sentence and feature level, with its main disadvantage being that it is highly domain oriented which means that words pertaining to one domain cannot be used in conducting sentiment analysis in another domain (Wankhade et al., 2022).

ML-based approaches involve the use of supervised machine learning (SML) and unsupervised machine learning (UML) models in conducting sentiment analysis tasks. SML algorithms need to be trained on a training set before being applied to new data, while UML discovers representative patterns and groups the data based on those (Wankhade et al., 2022).

When discussing SA on social media, papers Carvalho & Plastino (2021) and Wankhade et al. (2022) emphasize the importance of using as features the parts of text such as emojis, punctuation marks, and slang as these hold revealing information in respect to the

sentiment of the text analyzed. Extracting features such as term-frequency, tagging parts of speech and handling negations are other important considerations when using NLP techniques to determine sentiment or topics of the content analyzed (Wankhade et al., 2022).

Another aspect of processing text for NLP tasks is word embedding. This process groups words with similar meanings together and represents them in a vector space (Wankhade et al., 2022).

While SA provides a solution to analyzing text to understand opinions, impressions and reactions, the complexity of communication poses challenges to the NLP techniques used. Wankhade et al. (2022) and Zahoor & Rohilla (2020) discuss implicit language (sarcasm, irony, humor) detection as being one of the challenges of SA. Since implicit language is not represented in the text, the wrong polarity can be assigned when implicit language is not taken into consideration. To mitigate this risk, Fang et al. (2020) and Bouazizi & Ohtsuki, (2015) underline that analyzing emojis is an important aspect of detecting implicit language.

Models analyzing sentiment require computational power and a vast corpus to be trained on (Leeja & Bindu, 2020). This has led to further developments in the field of NLP and the release of pre-trained models with specific applications. Models such as VADER (Hutto & Gilbert, 2014) and BERTweet (Nguyen et al., 2020) deal with social media text, employing different techniques of handling features and text transformation.

To assess the accuracy of such models, the measures below were used in the literature. However, since these measures imply that the models need to be tested on the same dataset, Alsaeedi & Khan (2019) argue that in the absence of a benchmark, comparing SA pre-trained models proves to be difficult.

Recall	$\frac{True\ positive}{True\ positive + False\ negative}$
Precision	$\frac{True\ positive}{True\ positive + False\ positive}$
F1 Score	$\frac{2 * (Precison * Recall)}{Precison + Recall}$ A value closer to 1 on a scale between 0 and 1 indicates that the model has a good balance between precision and recall, hence accuracy.

Table 2 - Accuracy Measures

VADER is a lexicon-based pre-trained model. It has been developed through a combination of qualitative and quantitative methods to produce and validate a sentiment lexicon specifically tuned for microblog contexts (e.g. Twitter or other social media platforms) (Hutto & Gilbert, 2014). Hutto & Gilbert (2014) argue that VADER outperforms individual human raters in classifying sentiment of tweets into positive, neutral or negative, scoring an F1 value of 0.96 while human raters scored 0.84.

Alternatively, BERTweets is the successor of prior large models pre-trained for sentiment analysis such as RoBERTa and XLM-R and is the largest pre-trained model specifically on English tweets. When comparing the model to its predecessors, Nguyen et al. (2020) found that BERTweets performed better.

Research has covered extensively SA models in terms of the technical aspects behind them and their accuracy. Carvalho & Plastino (2021) evaluate the combinations of state-of-the-art features used in sentiment analysis and compare pre-trained models such as VADER, Sentiment140, Narr etc. based on these combinations. More specific studies such as Adwan et al. (2020) and Alsaeedi & Khan (2019) look into SA models and their performance on Twitter data. These showed that hybrid methods of ML classifiers and lexicon-based approaches performed well since they were able to benefit from the mix of approaches. However, Alsaeedi & Khan (2019) underline that determining the best approach for detecting sentiment on tweets can prove difficult since there is a lack of agreed benchmarks.

The development of pre-trained models has enriched the available tools to be used in conducting SA. Although, a multitude of papers exists on researchers training new SA models (Kohl et al., 2017) for the scope of this paper, a higher interest was assigned to papers where researchers have used pre-trained models as part of a toolbox in conducting studies related to SA. Numerous papers were conducted to understand perceptions or trends in relation to new technologies. Among others, studies covered topics such as trends related to perovskite solar cell technology (Lia et al., 2019), the acceptance of robotics (Sinha et al., 2017), the perception of the “Internet of Things" (IoT) (Bian et al., 2016), the adoption of Online Proctored Exams (Raman et al., 2021), the acceptance of blockchain technology for digital transactions (Grover et al., 2019), the adoption of AI and different aspects of chatbots (Borghi & Mariani, 2022; Cardona et al., 2019; Feine et al., 2019; Kushwaha et al., 2021; Sodré & Duarte, 2023; Widyaningrum et al., 2019).

For the scope of this literature review, the latter studies focusing on chatbots and AI have been offered particular attention. The similarities in methodological choices and theoretical frameworks between papers are worth mentioning as numerous of them make use of popular Twitter sentiment analysis (TSA) tools including the Natural Language Toolkit (NLTK) (Bian et al., 2016; Kushwaha et al., 2021; Lia et al., 2019; Sondré & Duarte, 2023) for data cleaning and text pre-processing, Latent Dirichlet Allocation (LDA) for topic modeling (Bian et al., 2016; Kushwaha et al., 2021; Lia et al., 2019) and VADER for sentiment analysis (Borghi & Mariani, 2022; Feine et al., 2019; Raman et al. 2021).

Another interesting commonality between the papers reviewed is how each of them has drawn upon similar existing theories. Based on the existing literature a common theoretical backbone related to technology acceptance and innovation can be observed. This backbone consists, among others of the following models: the Technology Acceptance Model (TAM) proposed by Davis (1989), the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003), Theory of Planned Behavior (TPB)

(Ajzen, 1985), Diffusion of Innovation Theory (DOI) (Rogers, 2003), and Information Systems Success Model (ISSM) (DeLone & McLean, (1992).

While all previously mentioned studies were taken into account, papers from Bian et al., (2016), Sinha et al. (2020), and Kushwaha et al. (2021) were considered of particular interest due to their relevance to the present study.

Bian et al. (2016) analyzed the perception over IoT by conducting TSA on tweets between 2009 and 2015. The study found that the general perception over the technology was positive, with tweets revealing concerns about privacy and security in relation to IoT (Bian et al., 2016). Sinha et al. (2020) draws theoretical implications from the TAM and UTAUT together with TSA to analyze the behavioral intention to accept robotics at workplace. The study identifies automorphism and technophobia as predictors of employee's intention to accept robots (Sinha et al., 2020).

Kushwaha et al. (2021) investigated the factors impacting customer experience of B2B companies using AI-powered chat-bots. In this inductive study, the research was conducted with the help of TSA and analyzed based on the theoretical implications of models such as Information System Success Model (ISSM) (DeLone & McLean, 1992) and DOI. The study concludes that the perceived risk and the information quality are significant factors for customer experience. If the information quality is a factor dependent to the algorithm, the risk is linked to users being more comfortable in discussing and providing personal information to humans (Kushwaha et al., 2021).

Based on the literature related to SA and social media commentary, it can be observed that some coding packages (e.g. NLTK, spaCy) and algorithms (e.g. LDA for topic modeling) are a common toolkit of methods and recommended to use for future studies. Additionally, a preference for drawing on the theoretical frameworks related to technology acceptance can be observed.

2.5. Concluding Remarks on Literature Review

In summary, the standing literature unveils that early adopters of GLLMs share a lot of enthusiasm and limited concerns. While the papers from different industries provide expert opinions on ChatGPT's impact for their respective industries, the studies from Haque et al. (2022) and Taecharungroj (2023) analyze tweets and therefore, depict opinions and sentiments from the masses.

Negative sentiments tie back to the possibility and easy access to produce complete academic tasks and projects, raising concerns around exam integrity and successful learning. A major concern also raised by Zarifhonarvar (2023) is shared by the general public. The issue deals with the hallucination effect, which is the confidence shown by GLLMs for wrong or incorrect answers. Haque et al. (2022) claim that it may impede the technology's wider adoptability. Positive sentiments link to summarizing and translating code, help with debugging, and create code in a fast manner, allowing to make people's jobs easier. Small teams will be able to develop incredibly useful products, services, and businesses thanks to AI coding and decision support systems (Haque et al., 2022). Other Twitter users from Haque et al. (2022) mention that compared to current search engines and knowledge-sharing platforms, ChatGPT works better in terms of search accuracy and speed.

Notwithstanding the fact that there are some similarities between the twitter analyses from Taecharungroj (2023) and Haque et al. (2022), this paper seeks to extend the literature by analyzing a larger corpus of tweets that also spans over a longer period of time. Additionally, the current status in literature does not examine Twitter user descriptions in detail to identify industries or groups of people and the sentiments connected to them. Besides that, existing studies could not scrutinize the development of sentiments over time, which is possible for this study as it spans over several months. Finally, this present study also expands on the current literature of analyzing social media commentary by employing TSA with the technical toolkits that already served other studies in the realm of SA. Albeit the first part of the literature review covers GLLMs, this section solely serves as a glimpse into their understanding and development of the

technology. However, this project adopts the perspective of a computational social science study.

3. Theoretical Framework

The present section serves as a theoretical backbone of this study. An introductory discussion of the general theoretical drawings related to technology acceptance, and a detailed explanation of the Diffusion of Innovation theory were considered essential to a study analyzing how the sentiment polarities of tweets may relate to the acceptance of the GLLM technology in different industries. For the scope of the present paper, the terms technology acceptance and technology adoption will be used interchangeably.

3.1. Fundamentals in Diffusion and Acceptance of Technology

According to Oye et al. (2012), technology is of little value until it is used. Technology acceptance, as defined by Louho et al. (2006), refers to how individuals accept and use technologies. Existing technology acceptance theories help researchers understand why some innovations or technologies are accepted more quickly by users and society than others. Next, a comprehension of the main models that have explained technology adoption in several studies will be given.

Existing models exhibit high versatility and they have been used in a variety of contexts since they are essential for evaluating and spotting new research trends of cutting-edge technologies like AI, augmented reality (AR) or blockchain (Lampo, 2023). Popular and widely used technology acceptance models in Information Systems literature are the Theory of Reasoned Action (TRA), the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), the Diffusion of Innovation theory (DOI) and the Information System Success Model (ISSM).

Technology Acceptance is a behavior, which results from a set of beliefs that people have about a technology in a logical, consistent, and frequent way. This idea is based in the Theory of Reasoned Action (TRA) by Fishbein & Ajzen (1975), which contends that the execution of a behavior is the result of the intention to perform that behavior. The theory

attempts to predict and understand almost any human behavior and has been applied in a number of fields. Two predictors, in turn, have an impact on the intention, namely the attitude and the subjective norm concerning the behavior in question.

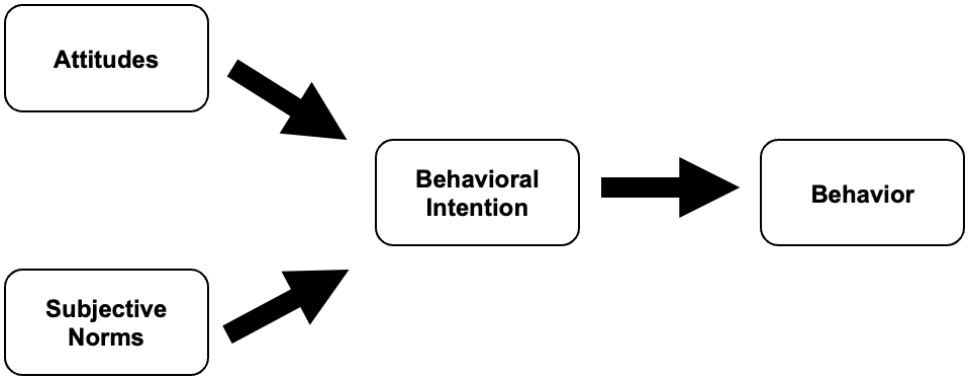

Figure 2 - Theory of Reasoned Action adapted after Fishbein & Ajzen (1975)

One of the adaptations of the TRA is the Technology Acceptance Model (TAM) which was originally proposed by Davis (1989). The TAM tries to explain user behavior across a wide range of computing technologies and user groups and to provide an explanation of the factors that influence computer acceptance. The perceived usefulness and perceived ease of use of the technology serve as the TAM's equivalent to the TRA's behavioral determinants. Ultimately, the UTAUT, which draws inspiration from the TRA, TAM and many more existing technology acceptance research, is a synthesized model that presents four key constructs (performance expectancy, effort expectancy, social influence, and facilitating conditions). Venkatesh et al.'s (2003) hypothesis states that these four constructs influence behavioral intention, which in turn predicts real use. Each construct has four moderating variables (age, gender, experience and voluntariness of use) and determines the degree of use of the technology in the end.

A study by Shuhidan (2019) analyzed the explained behavior variance, which is a crucial component of theoretical models. For example, the results of the paper show an explained variance in TRA (36%), TAM (52%), and UTAUT (69%). The higher the explained variance of a model, the higher is the ability to explain the variation in the data. However, according to Hagger (2019), there is a general recognition that theories can only provide partial explanations of behavior. According to Rogers (1962), acceptance is not always given albeit obvious technological advantages. In reality, new technologies fail to gain

widespread acceptance for a variety of reasons, for instance, users have become too comfortable in using their current technology and hesitant to use newer ones out of concern for making errors (Kim et al., 2016).

The Diffusion of Innovation theory claims that communication has a significant impact on social change within a community (Rogers, 1962). As the current research has its focus on attitudes and sentiment about GLLMs on the communication channel Twitter and less on the perceived usefulness, perceived ease of use or attributes of individuals, the authors decided the DOI social theory to be the most appropriate one to be employed in the analysis of this research, but not precluding the beforementioned theories in the analysis where additional theoretical drawings related to these make sense.

3.2. Diffusion of Innovation Theory

The Diffusion of Innovation (DOI) theory, which was developed by Rogers in 1962, is one of the oldest concepts in social science. It was first used to explain how an idea or product gains traction. As a result of traction and diffusion, people eventually accept a new concept, behavior, or innovation as a part of a social system. The theory has been reviewed over time and the focus shifted towards the adoption of new technologies. Adoption depends heavily on perception and the main goal of the DOI theory is to explain how any technological innovation moves from the stage of invention to widespread use (Dillon & Morris, 1996).

One of the key concepts of DOI is that diffusion happens in stages and moves through different groups of adopters. Some people are more likely to embrace the new technology than others and adoption does not happen all at once but gradually. According to Rogers (1962), adoption of a new technology follows a bell curve. Moreover, depicted in Figure 3, Rogers established five categories of adopters with different traits, which are crucial to understand, facilitate or impede adoption.

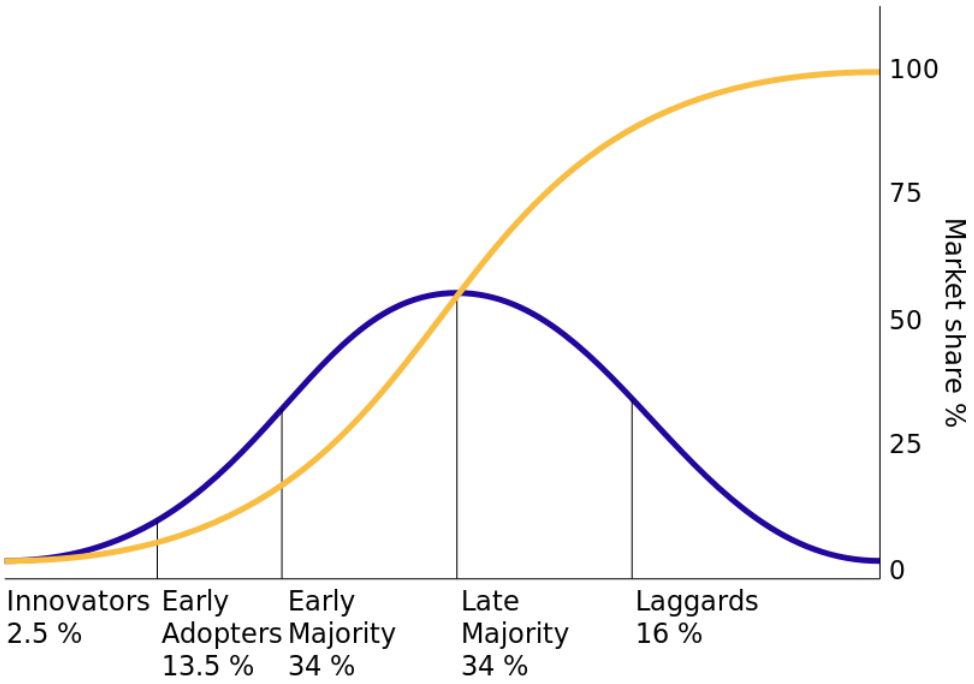

Figure 3 - The Five Categories of Adopters and the Adoption Curve (Rogers, 2003) adapted by Wikipedia (2013)

Innovators are most open to change, and laggards are the most hesitant to probe new technologies. Apart from the innovators, every category adopts an innovation by observing experiences from the previous reference group. Adoption grows throughout the population as positive perceptions and testimonies accumulate. The adoption of an innovation by 10–20% of the social system's members marks a critical take-off moment. At this time, the rate of diffusion is at its maximum and a growing number of people are accepting the innovation. It is often impossible to halt further diffusion until market saturation after the take-off point.

Innovators

Individuals, that are eager to test out new ideas first. They are curious about novel concepts and first come up with ideas for further adoption. Ultimately, innovators are crucial to the diffusion process as their actions often launch the spread into society.

Early Adopters

Individuals, who are thought leaders in their fields. They appreciate taking on leadership responsibilities and embracing change. They are open to implementing new concepts because they are already conscious of the need for change.

Early Majority

Though they hardly ever take the lead, members of this group are often the first to embrace novel concepts. Before they are willing to adopt an innovation, they usually require proof that the new technology is effective.

Late Majority

These people are resistant to change and won't embrace a new idea until the majority of a social system has established the innovation.

Laggards

These individuals are very traditional and the toughest group to convince to accept change because they are so resistant to it. Individuals in this group exhibit little to no opinion leadership. In contrast to some of the prior categories and laggards might not adopt the new technology at all.

Furthermore, diffusion is defined as the process by which an innovation is communicated through certain channels over time among the members of a social system (Rogers, 2003). The four main elements (innovation, communication channels, time, and the social system) are further described as follows:

Innovation

An innovation is an idea, a concept, or an object that someone perceives as new. If someone is aware about an innovation for some time but has not yet developed a favorable or unfavorable attitude towards it, the individual has not adopted or rejected it (Rogers, 2003). The adoption of an innovation is also tied to the following five perceived attributes of an innovation: relative advantage, compatibility, complexity, trialability, and

observability (Rogers, 2003). The degree to which an innovation is seen as superior to the concept it replaces is known as relative advantage. Compatibility entails how an innovation is perceived as consistent with the current values and prior experiences. The perception of ease of understanding and applying an innovation is referred to as complexity. The dimension of trialability deals with the accessibility of an innovation to be tested out. And finally, the visibility of the impact and benefits of an innovation contribute to the observability.

Communication Channel

The second main element of the DOI is communication channels. According to Rogers (2003), a message created and shared between individuals with a new idea can be described as diffusion and a communication channel is the means for two people to communicate with one another. Albeit traditional mass media, e.g., radio and tv, and interpersonal communication channels, such as face-to-face exchanges, are named as communication channels, Rogers also points out that interactive communication over the internet has grown in significance for the spread of innovations. Moreover, he states that most individuals fundamentally depend upon a subjective evaluation of an innovation that is relayed to them from peers who have already adopted the innovation. This indicates that the core of the diffusion process involves imitating their network partners and their perspectives.

Time

The time dimension revolves around the innovation-decision process, which describes the steps an individual takes from first learning about an innovation to developing an attitude toward it, to deciding whether to adopt or reject it, to putting the new idea into practice, and to having this decision confirmed.

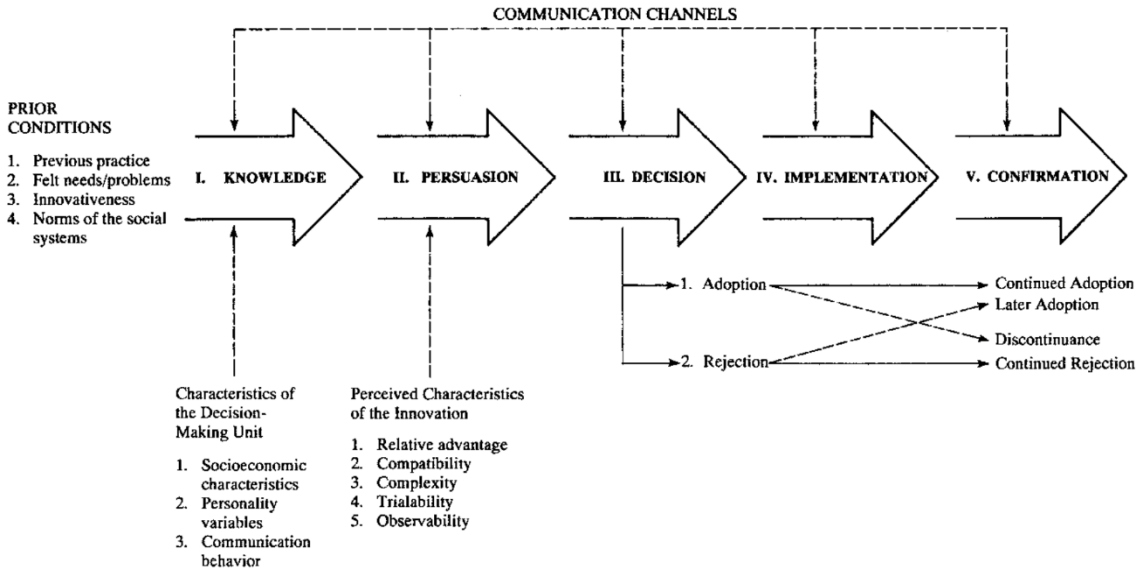

Figure 4 - Innovation Decision-making Process, based on Diffusion of Innovations (Rogers, 2003)

The essential steps in the decision-making process, i.e., adoption process, are:

- **Knowledge**: When a person is introduced to an innovation and learns how it works.
- **Persuasion**: When a person develops a positive or negative attitude toward innovation, persuasion takes place.
- **Decision**: A decision is made when a person makes actions that result in a decision about whether to accept or reject the innovation.
- **Implementation**: When a person puts a new idea into use.
- **Confirmation**: An individual wants confirmation of an innovation-decision they have already made, but they may change their minds if they encounter contradictory information about the innovation.

Social System

Diffusion takes place within a social system, which may be individuals, groups or organizations. Influential people can spearhead resistance or promote new ideas. With Twitter being the social media platform where opinion leaders and users from across the world share their thoughts (O'Malley, 2022), the authors consider Twitter to be the social system in an influential position for society as a whole. Another consideration derived from considering Twitter the social system is that topics become the sub-systems in this

case. Members of a social system can facilitate or impede adoption depending on the acceptance of an innovation to its culture. The Diffusion of Innovation helped researchers explain the spread of new technologies in a range of fields, such as in health care with the X-Ray technology or in consumer goods with the personal computer (Sirk, 2020). To conclude, the authors believe that the adoption of GLLMs can be analyzed with this theory and that TSA could be related to the two stages of the DOI process (knowledge and persuasion). While the mental activity was cognitive thinking at the knowledge stage, a feeling is the main type of thinking in the persuasion stage (Rogers, 2003). It is logical as one cannot begin to develop an attitude toward a new idea until one is aware of it. To advance the understanding of Twitter users' acceptance of GLLMs, elements of the DOI will be used as the guiding theoretical lens.

4. Methodology

The present chapter discusses the methodological choices made in relation to this study with the scope of answering the explanatory research question: How are GLLMs perceived by participants of different industries based on social media commentary?

To expand on the layers of the present research, the research onion model was used (Saunders et al., 2019). The research onion is one of the most commonly used models in organizing the methodology aspects of a research paper as it provides a facile framework delimiting between all the following research aspects: philosophy, approaches, strategies, choices, time horizons, and techniques and procedures.

The first section talks about the research's general philosophy, which will be followed by the research strategy. Subsequently, the methodological choice and the methods, including the data collection and data analysis will be covered.

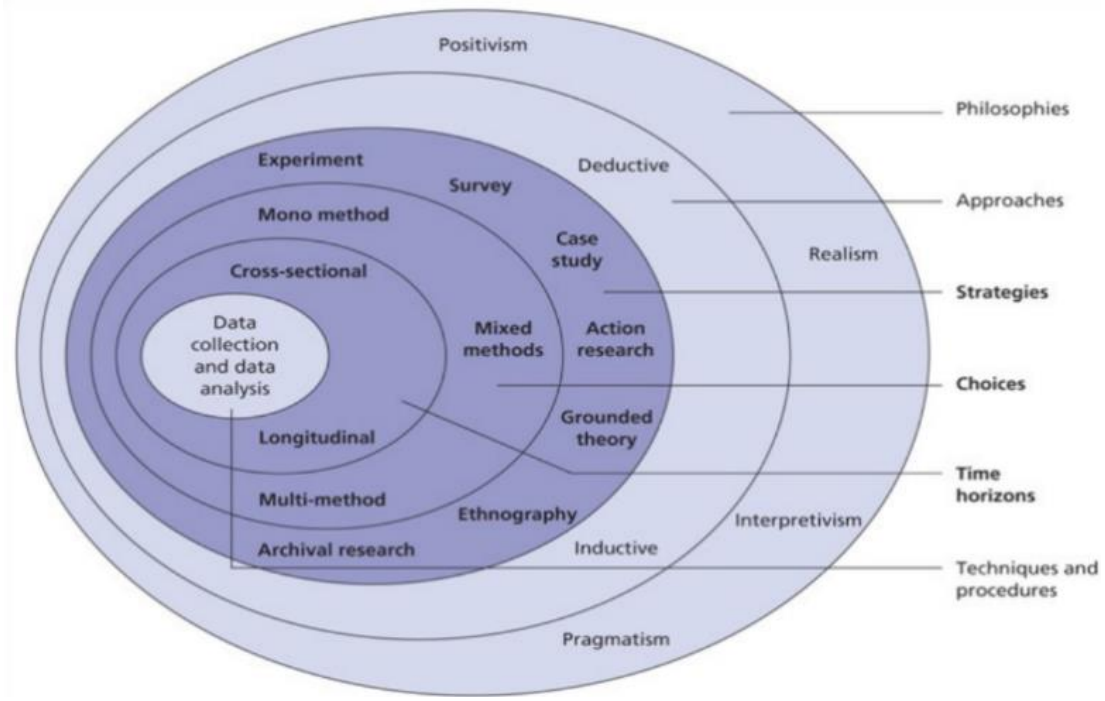

Figure 5 - Research Onion adapted after Saunders (2009)

4.1. Research Philosophy

This study adheres to the pragmatism research philosophy as it seeks to derive new knowledge based on methods considered most appropriate (Saunders et al., 2019). The affiliation to pragmatism can be denoted based on the object of the study: the perception of GLLMs of different industry participants. With the perception of a new technology being the focal point, no knowledge or method can be unambiguously considered. Cognizant of this, the present contemporary study combines elements of quantitative and qualitative methods most appropriate to achieve greater validity, reliability and representativeness in studying the perception of GLLMs and answering the research question.

The pragmatist point of view recognizes that there are many different ways of interpreting the world and that no single philosophical perspective can provide a holistic view of a study (Saunders et al., 2019). Hence, positivist and interpretivist methods are used together. Quantitative data analysis methods and the use of algorithms underline the positivist aspect of the study. On the other hand, the study also adheres to interpretivist ideas, as the findings are analyzed through the lenses of the theoretical

drawings related to technology acceptance, aiming to capture meanings in human interaction (Black, 2006).

4.2. Research Approach

The research approach used in the present study is an inductive one. This approach entails that the research starts from specific observations, which are then studied to draw conclusions or generalizations (Saunders et al., 2019). In the present research, the specific observations are represented by the social media commentary (tweets) on ChatGPT. The characteristics of the specific observations are what strengthen induction as being the approach used in the study. The data used is specific since it is a sample from the existing social commentary on ChatGPT. The use of ChatGPT as a proxy for GLLMs, in turn, underlies the fact that generalizations will be made based on specific observations.

4.3. Research Strategy

The research strategy intends to expose how the authors plan to answer the research questions and reach the objectives of this project (Saunders et al., 2009). For this study, the research strategy revolves around computational social science and netnography that should justify the overall approach and the individual methods to be used in the study.

4.3.1. Computational Social Science

According to Wallach (2018), analyzing Twitter posts with DOI falls under the category of computational social science (CSS). Wallach described CSS as the study of social phenomena through the use of digital data and computational and statistical techniques and sits at the intersection of computer science, statistics, and social science. CSS is acknowledging that machine learning alone with algorithmic black boxes is not the answer to all. Researchers must think outside the box when it comes to opportunities and challenges using ML methods to analyze real-world data about society. The authors try to follow this appeal by analyzing the adoption of GLLMs through the combination of

exploring social media commentary on Twitter with ML methods and the social theory of DOI.

4.3.2. Netnography

Netnography is an adaptation of ethnography that utilizes ethnographic research techniques optimized for the internet. For the study of online groups, networks, and social media, netnography is a research methodology that is more naturalistic and unobtrusive than focus groups or interviews, as well as quicker, easier, and less expensive than conventional ethnography (Kozinets, 2002). Netnography entails methods of extraction, exploration, and sentiment analysis for computer-mediated conversations to derive cultural understanding in any given context. Researchers using netnography combine qualitative and quantitative techniques to derive customer attitudes and comprehend the mindset of customers in the virtual world.

A vital point for the authors was the matter of research ethics with netnography. Netnography deviates from conventional research in such a way that the use of cultural data is not deliberately and confidentially provided to the researcher (Kozinets, 2002). The authors believe that one of the most fundamental and universal ethical concepts is the dignity of people and the protection of their personal information. Sudweeks & Rafaeli (1996) formed the ethical policy that the issue of informed consent of participants is not required when publicly available text is analyzed. Generally, informed consent is not required if you may access the data without logging in and people would expect to be observed by strangers, such as on a platform like Twitter (Shaw, 2018). According to Shaw (2018), one caveat to researching digital platforms is that the researched data must remain anonymous. Only then, research can be done without getting permission and informed consent from the posters.

In the present study, the object of analysis is the communication on Twitter that is openly posted by users. However, Twitter users were anonymized, thus never named or quoted. The authors would like to state that certain precautions were taken to provide anonymity to the users examined for this research. The applied research techniques were considered carefully to ensure that ethical and candid netnography is conducted.

4.4. Methodological Choice

This research can be seen as methodologically innovative because it tries to explain large-scale digitized data sets with a sociological lens. For that reason, the theory-in and theory-out framework developed by Radford & Joseph (2020) should serve as guidance and inspiration. Their study scrutinized the relevancy of connecting the application of machine learning tools and their results to social theory (Radford & Joseph, 2020). The "theory-in" part refers to the process of conducting the technical steps with the respective social theory in mind. This involves the selection of relevant data and methods to get the outcomes that can be interpreted in the “theory-out” part. The "theory-out" part implies using social theory to make sense of the results of the ML methods.

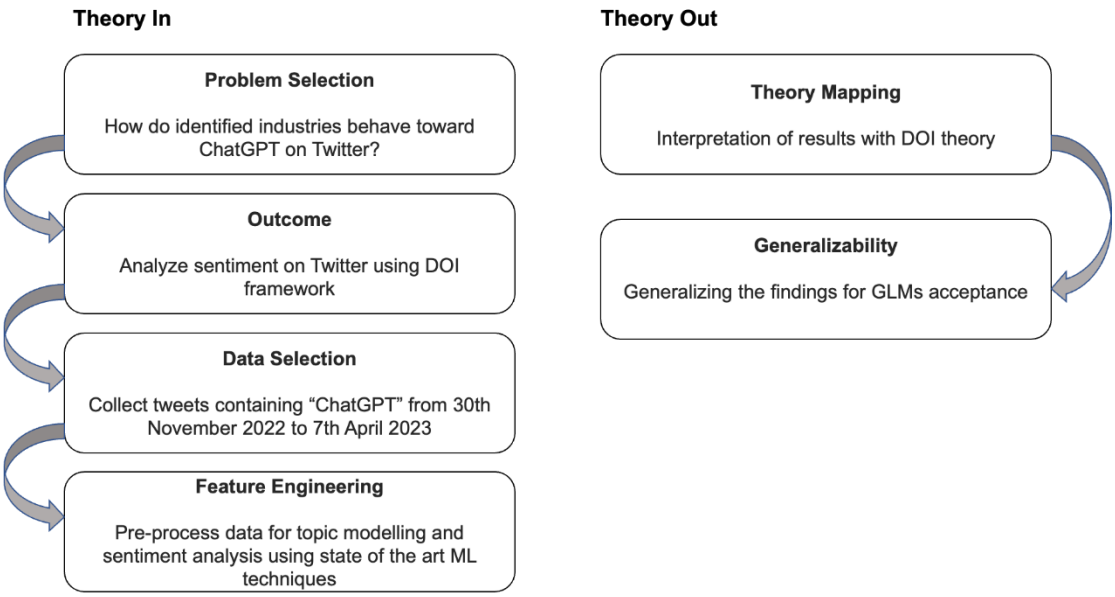

Figure 6 - Research Methodology Process adapted from Radford & Joseph (2020)

Figure 6 showcases the overall research methodology, which is the theory-in and theory-out framework (Radford & Joseph, 2020) adapted to the scope of this research. A telling example to underline this approach regarding validity and reliability would be the study by Al-Razgan et al. (2021). They made use of the theory-in and theory-out framework by Radford & Joseph (2020) to employ Rogers’ (2003) innovation-decision process as a lens for the analysis of their Twitter sentiment findings. Similar to Al-Razgan et al. (2021), the authors have mapped the theory-in part to sentiment analysis and the theory-out part was attributed to the DOI theory.

Furthermore, this present study can be described as a multi-method approach. Although there is only a singular data collection method (Twitter scraping), the extracted qualitative data (tweets) is further analyzed through exploratory data analysis, topic modeling, and sentiment analysis. These analysis steps are in turn generating quantitative data such as the number of tweets, the percentages of positive and negative tweets, and more. To summarize, the main objective of applying multiple methods is to examine what information about GLLMs is shared, what can be construed out of that data and how can it be given sense through the DOI theory.

4.5. General Assumptions and Considerations

In this section, the authors want to disclose and gather the assumptions that played a significant role for making decisions about the methods applied for data collection and data analysis.

To start with, the authors assume that focusing on tweets about ChatGPT is sufficient to find substantial results for answering the research question. Statista (2023a) shows that the surge in interest about generative AI concurs with the launch of ChatGPT. Other keywords, such as “large language models” or “generative language models” were considered to be included in the scope of scraping. However, the number of tweets extracted were insignificant compared to the data set that resulted from only scraping tweets that included ChatGPT. Often times, ChatGPT showed up along these, as well. To avoid dealing with potential duplicate tweets on large datasets and juggling several keywords that do not benefit the study vastly, the authors inferred to go ahead with this assumption for this project.

Secondly, the authors presume that topic modeling on user descriptions will deliver meaningful results to classify the respective user to an industry. By observation, the authors identified that users often link their domain of their professional activity in the user description. Nevertheless, the user description field is not restricted to any specific professions or industries but could contain any kind of text within the limit of 160 characters. Although it is possible on Twitter to turn a user account into a professional account and choose a category with the given picklist, it was not feasible to extract this

specific data. Therefore, the authors decided to use unsupervised machine learning to investigate whether significant information related to industries can be uncovered with the available user description field.

At last, the DOI theory as a theoretical lens needs to be discussed. After the machine learning techniques of topic modeling and sentiment analysis uncover more insights on the industries and their sentiments, the authors are desirous to interpret these results with the DOI theory by Rogers (2003). Koçak et al. (2013) have already compared their findings of interview results on social media usage with the five stages of the decision-making process. Another compelling instance is Al-Razgan et al. (2021), who based their data analysis on mapping attitudes towards driving adoption by Saudi women, obtained through tweets, to the five stages of the decision-making process. Both studies treat Rogers' framework as a tool for interpretation, and to further rationalize their results meaningfully.

The authors contend that the diffusion process with its four main elements (innovation, communication channels, time and social system) are identifiable in the present study. The examined innovation is ChatGPT, a public GLLM that is communicated on the communication channel and social system of Twitter. This society's tweet sentiment serves as a ground to make assumptions about the diffusion of ChatGPT, whether this technology gets accepted or rejected, and how the further adoption process in this social system might look like.

The authors inspect Twitter as a society, which is a social network with around 350 million active users (Statista, 2022). According to Rogers (2003), if a user has not yet developed a **favorable** or **unfavorable** sentiment towards an innovation, the individual has not adopted or rejected it. The authors claim that the extracted sample of Twitter society has gained knowledge about the technology and thus, is able to exchange information about it. Hence, every user in the data set has passed the knowledge stage of the decision-making process and stands in the persuasion and decision stage. Users that have not tweeted about ChatGPT and therefore, have not developed any tangible sentiment yet, will be categorized as the part of the adoption curve, that have not adopted the innovation.

The sentiment analysis results will disclose how users, that have gained knowledge about ChatGPT, feel towards this innovation. Users at this stage stand at a vital decision-making point that could influence the further diffusion process. The authors assume that the sentiment results can be classified like the following:

Positive sentiment: Indicator that user has tried, seen or read about ChatGPT and has a positive opinion. This user is likely to take action with the attitude to adopt this new technology or continue further use.

Neutral sentiment: Indicator that user has read, seen or tried ChatGPT, but is not feeling convinced yet to either accept or reject this innovation.

Negative sentiment: Indicating that user has tried, seen or read about ChatGPT and cope with bad feelings or negative experiences. The user will take decisions to act with the attitude to reject or adopt the new technology at a later stage.

Regarding the implementation and confirmation stages that come after the decision phase, the authors consider that sentiment data alone cannot make convincing statements. Hence, the authors will refrain from making any assumptions and interpretation on these stages. The authors suggest that further qualitative research, for instance, through interviews or manual examination of tweets, needs to be conducted to get substantial results for the latter mentioned stages.

4.6. Data Collection and Processing

With the scope of answering the research question of this study, the data collected is qualitative and consists of tweets about "ChatGPT". The choice of tweets was based on the fact that this project considers "ChatGPT" a proxy for GLLMs. Thus, the information stored in tweets about ChatGPT reflects the opinions about GLLMs. The tweets scraped were published between 30th of November 2022 and 07th of April, underlying the longitudinal character of this study.

Twitter data is appropriate as users communicate their views and impressions in a short text by using particular language, tones, and editing behaviors, revealing opinions.

Combining insights through social media data with a sociology theory allows the authors to focus not only on the attitude over the technology being positive, negative, or natural but also on how the observations translate to changes in society.

Based on the above the authors identified the sub-scopes of collecting and analyzing the data prior to mapping their findings to the theoretical framework chosen, and employed the below-mentioned methods:

Scope	Method
Data collection	Web scraping with the help of the twitter module of snscrape, a python package allowing for social media data extraction (JustAnotherArchivist, n.d.).
Identify user industry	Topic modeling based on Latent Dirichlet Allocation (LDA), an unsupervised generative statistical topic modeling model (Blei et al., 2003).
Determine sentiment classification of each tweet	Polarity Classification performed by Valence Aware Dictionary and Sentiment Reasoner (VADER), a lexicon and rule-based sentiment analysis tool that was specifically tuned to cater to the analysis of sentiments expressed in social media text (Hutto et al., 2014).
Identify common topics of discussion in collected tweets	Topic modeling based on LDA.
Analyze visually the collected data	Visualization techniques supported by python packages such as matplotlib (Hunter, 2007) and altair (Sievert, 2018), combined with PowerBI (PowerBI, 2023) generated visuals.

Table 3 - Technical Process of Data Collection and Analysis

The method chosen for data collection in the present research was web scraping.

The choice of collection method stands in the limitations of Twitter's application programming interface (API). According to Twitter, through the company's API endpoints, you can access Tweets, users, spaces, private messages, groups, and more. To get academic research access and the current version of the Twitter API, a Twitter developer user account is needed, which the authors applied for on 7th of February 2023. The access was, however, never received due to long processing times from Twitter's side. Based on the company's website it can be concluded that Twitter is working on new developments of the Twitter API (Appendix 1) (Twitter, 2023).

To avoid these restrictions, and adhere to the timeline of the project, the authors sought alternative methods for data collection. Thus, the authors used the open-source python scraper package "snscrape" (JustAnotherArchivist, n.d.), which provides the functionalities

needed for collecting historical tweets and additional information about these, such as date of posting, number of retweets and more, hashtags.

The methods deployed in topic modeling and polarity classification were chosen based on their documented effectiveness and popularity in the existing literature.

When choosing the topic modeling algorithm, the decision was largely made from standardization purposes and due to other research papers handling similar topics with the chosen algorithm. Suri & Nihar (2017) and Khan & Chua (2021) concluded that LDA outperformed models like Non-Negative Matrix Factorization (NMF) and Latent Semantic Analysis (LSA) respectively. On the flip side, Egger & Yu (2022) state that LDA's performance is similar to the one of models like Top2Vec or NMF and that it is underperforming newer models such as BERTweet. Despite these conclusions, the paper goes on to underline that LDA finds coherent topics when its hyperparameters are tuned and that since it usually generates a smaller number of topics than word embeddings, these are easily interpretable. As a result, LDA was employed for topic modeling in the present study. The model's performance was then assessed based on coherence scores. The results of the performance assessment were then used to hyper-tune the model.

As will be further discussed, the results of the topic modeling on tweets content were considered not cohesive. Thus, to further enrich the data, the authors of this paper investigated keyword extraction techniques as complementing exploratory methods.

VADER is a lexicon and rule-based sentiment analysis tool. It can efficiently grasp vocabulary, abbreviations, capitalizations, repeated punctuations, and emoticons typically found in social media posts to express an opinion or idea (Hutto & Gilbert, 2014). According to Bonta & Janardhan (2019), VADER's advantages are that it works great on social media text, it does not require any training data and it supports emojis in sentiment classifications. Together with the fact that VADER does not impose a major speed-performance trade-off, these were some of the favorable arguments for employing VADER in the present analysis. Additionally, based on the experimental approach of Bonta & Janardhan (2019), VADER outperformed other polarity classification tools such

as Textblob and NLTK. This information supports Hutto et al.'s (2014) findings, which state that VADER has an F1 classification accuracy of 0.96.

In the following parts of this section, the main steps of this research will be described.

4.6.1. Data Collection & Anonymization

To be able to ensure the collected data is rich enough for the scope of this project, finding the variables that can be used to scrape Twitter was necessary before collecting any data. To keep the foundational data for this study as broad as possible, the scraping was conducted on tweets that contain "ChatGPT" in their body or hashtags. The choice to query solely based on "ChatGPT" stands in the desire to gather a large dataset of tweets and information about the authors of these tweets, based on which further analysis can be conducted to identify especially relevant datapoints. Broadening the keywords search to terms such as "large language models", "generative language models" or "generative artificial intelligence", was considered. However, initial observations made by the authors led to the conclusion that the two firstly mentioned terms were not as popularly used as "ChatGPT", and that in the few instances where they were used, they appeared along "ChatGPT". Additionally, it was considered that certain terms such as "generative artificial intelligence" would result in data covering a broader scope than the one of this project.

The data collection process, as documented in the "Data Collection" section of Appendix 2, was done with the help of a python *data_collection* function which stored the scraped data into CSV *(comma-separated value)* files. Due to computational limitations the scraping process was conducted in increments that have resulted in nine csv files. Due to ethical considerations of maintaining the privacy of Twitter users, the raw data of this project will not be uploaded, instead, data sets consisting of the anonymized raw data will be provided.

The raw data has 19 columns, containing tweet information, such as tweet id, date of publication, content, username, user profile description, and additional metadata, e.g., tweet vibe, user verification status, and number of retweets that could potentially be used for further analysis. Secondly, the data frame was saved as a CSV file. The final merged

dataset comprises 1 640 146 tweets about ChatGPT, written in English, between 30th of November, when ChatGPT was released, until the 7th of April 2023 inclusively. Throughout the data cleaning and data preparation processes, 100 tweets were discarded due to the inability to process them. This resulted from the 100 tweets being recorded in the wrong formats that were not processed by the helper functions used in this project. Therefore, the number of tweets taken into consideration for the present study is 1 640 046 tweets that do not contain media content, pertaining to 698 701 unique users.

Once data collection was complete, the data was cleaned for anonymization purposes. The raw data stored Twitter accounts' information (Twitter usernames) in three columns: "usr_userid", "usr_username", and "usr_inreply". The first column refers to the id of the author of the tweet while the second refers to their user handle. The third column stores the user handle of the author of the tweet to which the tweet replies. While the user handle of the author of the tweet was easily anonymized by removing the usr_username column, another method had to be used for the usr_inreply column.

Thus, a dataset was created from all the unique user IDs and their Twitter account handles and stored in a CSV file "User_Map.csv". The user handles stored in usr_inreply column was then mapped against the new dataset and replaced with a user id. For the users that could not be found in the new dataset, these were replaced with "nan" values.

Additionally, to the anonymization process, the data was cleaned by the removal of columns that were deemed redundant: ("Tweet Vibe" and "Tweet Media"). Tweet Vibe stored information for a very small number of observations and tweet media was redundant as the present paper does not investigate tweets containing media.

As a following step, duplicated tweets were removed. These were recorded throughout the scraping process and are assumed to be a result of page refreshes.

A final step in the initial data cleaning was the split of the main dataset into two smaller ones, each serving a different purpose in the following steps of the analysis. Thus, user ids and user descriptions were stored in the UsrData_PreLDA.csv file, while tweet information (tweet_id and tweet_content) was stored in Tweets_PreLDA.csv. The user data was then used for determining if users are connected to an industry and if so to

which one, while tweets data was used to determine their sentiment polarity and what topics were discussed in relation to ChatGPT.

4.6.2. Pre-Processing for Topic Modeling

Instead of manually labelling the industry to which users belong and the sentiment of each tweet, ML methods were used to automate these tasks. Using ML algorithms makes it feasible to handle large datasets and circumvent manually coding qualitative content. As a prior step to the text analysis with ML, pre-processing was conducted. This is the procedure of using data cleaning methods to format the data to the input requirements of the ML algorithms later used. This step is critical as pre-processing improves the accuracy, efficiency, and consistency (Mhatre et al., 2017) of the output of ML algorithms by increasing the input data quality.

To effectively use LDA for topic modelling, certain data cleaning considerations needed to be performed. These considerations are grounded in the model operating on a bag-of-words. This implies that the model places importance on the frequency of words and not on their order in the sentence. As a result, to ensure the quality of the input data, it is important that only words that are relevant to the topics existing in a text are considered in the bag-of-words.

Since topic modeling will be performed both on user descriptions and tweet contents, a preprocessing function was created to prepare each of these texts as bags-of-words.

The function transforms the text through the following steps:

1. Extraction of hashtags (#), mentions (@username), emoticons, hyperlinks, and URLs in separate columns, since this information may be part of the latter analysis.

2. Cleaning, i.e., removing of words and special characters including mentions, URLs, emoticons, hashtags, and numbers from the original because they are deemed redundant for determining the overall subject in a bag of words.

3. Removal of punctuation as these are not required to model topics.

6. Tokenization, which resulted in breaking sentences into words. For this step, the "spaCy" package was utilized, as it is a commonly used, open-source, python library for pre-processing in NLP use cases. This library is not only capable of segmenting sentences into words but also to understand word relationships and part-of-speech tagging, which is assigning word types to tokens, e.g. verbs or nouns (spaCy, 2023).

7. Lemmatization, as this is a crucial pre-processing step for text mining applications and can be described as the process of finding the normalized form of a word (Plisson et al., 2004). To perform this step, the "LemmInflect" package was used. The imported methods change the words found in the initial field into their root forms. For instance, "buying" becomes "buy", "meeting" becomes "meet", while comparative forms such as "better" and "best" are changed to their ground form, e.g. "good". This has the benefit of lowering the overall number of unique terms and consequently, assisting the machine learning classifier used for topic modeling in producing better topics.

8. Stop-words removal as this step strips the initial text of extremely common words that are considered to have no message and to hold no meaning in the sentence. As these words usually appear in large numbers the topic produced would not be interpreted properly if this step is not taken. English stop-words as provided by the "NLTK" package were removed. The list of words that do not contain relevant information for the text classification are prepositions, conjunctions, pronouns, and articles such as "he," "she," "a" and "the". After an interim analysis of topics generated, additional words that turned out as noise, e.g., "et", "also" or "more", were removed.

In the case of cleaning the tweet content, an extra step was added of stripping tweets of verbs. It was considered by the authors that verbs may deter the results of topic modeling as opinion verbs such as "think", and "believe" do not provide additional information about topics. However, this additional step was added after the initial LDA results on tweets topics were conducted, and the consideration that verb removal may also be detrimental to topic modeling was also made. Thus, a new column where verbs are excluded from was created. LDA will be thus conducted on both columns separately and the outcomes of each model will be compared.

Column	Data Description
Usr_userid	Unique user-ids.
Usr_description	User description scraped.
Usr_description_mentions	Accounts are mentioned in the user description.
Usr_description_hashtags	Hashtags extracted from the user description.
Usr_description_URLs	URLs extracted from the user description.
Usr_description_emoji	Emojis extracted from the user description.
Usr_description_clean	User descriptions were decapitalized and with digits, punctuation marks, stop-words, URLs, emojis, and mentions removed.
Usr_description_tokens	List of lemmatized tokens (words) forming the user description.

Table 4 - User-specific Data

Column	Data Description
Tweet_id	Unique tweet-ids.
Tweet_content	Tweet content.
Tweet_content_hashtags	Hashtags extracted from the tweet content.
Tweet_content_mentions	Mentions extracted from the tweet content.
Tweet_content_URLs	URLs extracted from the tweet content.
Tweet_content_emoji	Emojis extracted from the tweet content.
Tweet_content_clean	Tweet content decapitalized, and with digits, punctuation marks, stop-words, URLs, emojis, mentions removed.
Tweet_content_tokens	List of lemmatized tokens (words) forming the user description.

Table 5 - Tweet-specific Data

The objective of this data pre-processing stage was to prepare the data for topic modeling by splitting the sentences into rooted (lemmatized) tokens (words) and removing elements of noise such as digits, mentions, URLs, and more.

All 1 640 046 tweets were tokenized successfully and were taken into consideration for both sentiment analysis and topic modeling.

4.6.3. Topic Modeling

After the initial data pre-processing, topic modeling on user descriptions was performed. Topic modeling is an ML task that can be performed with the use of statistical algorithms that are used to extract underlying themes or latent subjects from big unstructured collections of texts (Blei et al., 2003). In this study, the collection of user descriptions is categorized into latent topics using LDA. LDA has been demonstrated to be a successful unsupervised machine learning technique for uncovering implicit document topics (Blei et al., 2003; Chen et al., 2017) and, as described in section 2.4., it has proven to be popular among papers researching the relationship of different concepts (e.g. IoT, robotics at workplace) and social media.

The performance of LDA on the user description data was done based on the coherence scores recorded.

In the context of LDA, coherence is a measure of the interpretability of the topics generated by the model. The measure calculates the degree of semantic similarity between the highest-ranked words of each topic. While coherence alone does not guarantee quality in the generated topics (Rosner et al., 2014; Nikolenko et al., 2015), it is a good starting point in assessing an LDA model. The coherence score can take values between 0 and 1, with values above 0.45 to 0.7 being considered good and realistic among the data science community (Stackoverflow, 2019). The coherence score of an LDA model is influenced by its parameters, such as the number of topics, alpha, and beta (eta). With alpha showing whether a document, or tweet in this context, is likely to contain multiple or fewer topics. A lower alpha underlines that each tweet would contain a small number of topics. The beta parameter determines whether a topic consists of a large or small number of words. A low beta implies that topics are sparse, with few words that have higher probabilities to belong to the specific topic.

When hyper tuning the LDA models used in this study the hyperparameters used differed based on the coherence threshold set by the authors and on the scope of the LDA. The coherence scores deemed acceptable for the authors of this paper needed to be above 0.5, with higher values being considered improvements.

When discussing the topic modeling procedures used in this paper, the following common phases can be identified:

A. The data cleaning phase: where newly identified redundant words were removed from the user descriptions.
B. The base implementation phase: when an LDA base model was fitted to the user descriptions.
C. The hyper-tuning phase: when a better fitting LDA model was picked based on performance which was evaluated with the help of coherence scores.
D. The labeling phase: where users were categorized in one of the industries identified by the model chosen in phase C.

For both topic modeling scopes (industry identification and tweets' topics of discussion identification) the LDAMulticore module of the "genism" python package was used.

It is worth mentioning that despite the commonalities between the phases of LDA, these include different applications based on the scope of topic modeling.

For the LDA conducted on users' descriptions, once user data was loaded it was observed that a few other words/strings had to be removed. These included the 'nan' string, a string consisting of an empty space, and one-character strings (e.g., 'a','e').

This was followed by phase B which resulted in declaring an LDA base model with 12 topics. Derived from the visualization of the LDA-based model, additional words were removed as documented in the first section of Appendix 4.

After the final data cleaning, 593 657 user descriptions were identified as viable for topic labeling. The remaining users had their descriptions empty or consisting of standalone URLs, emojis, or special characters. These descriptions were not considered for topic modeling as they resulted in no tokens.

In phase C, calculated the coherence score of models fit with the number of topics between 6 to 20 in steps 2. Since the coherence score results were considered acceptable, no further tuning of the parameters alpha and beta was required.

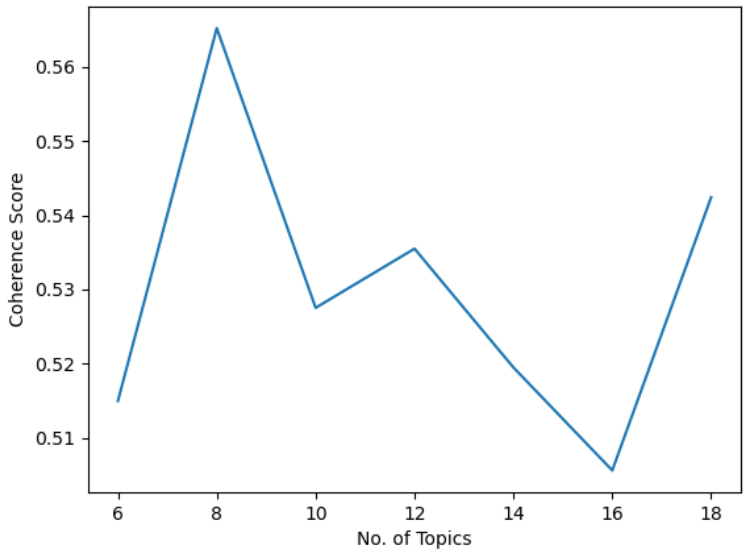

Figure 7 - LDA Coherence Score by Number of Topics in steps of 2

The LDA model generating 8 topics recorded the highest coherence (0.565) and thus was used to label the user descriptions into industries.

Based on the first 10 words pertaining to each identified topic, the user descriptions were labeled to an industry category.

The initial phase A of tweets' topic labeling was very similar to the one of industry identification. However, apart from certain one-letter strings being removed, here words such as "chatgpt", "ai", and "artificial", "intelligence" was removed as they were considered to be redundant for the topics of discussion since all tweets are centered around the main topic defined by these.

In phase B, a base LDA model was trained with 10 topics for each of the columns storing tweet tokens (column including verbs and excluding verbs). This resulted in a coherence score lower than 0.5 for both cases, which was not considered viable. To increase the coherence of the model, the top 1000 words belonging to the corpus were analyzed. As a result, phase A was repeated by removing more words such as "mostly", "especially",

and “unlikely” and more as detailed in Appendix 5. These words were considered filling words and not words that hold information about a specific topic.

Subsequently, phase C hyper-tuned the model based on both token columns (column of tokes including verbs and column excluding verbs) on multiple values of several topic numbers, iterations, and alphas. Since the base LDA offered a preview of some topics that may be uncovered in the data, the authors decided to train LDA models with different alphas.

The LDA models recording the highest coherence scores were the ones trained with asymmetric alpha values, for 100 iterations on tweet tokens including verbs. Upon inspecting the topics of the highest-scoring model, the authors of the study observed that the topics are not as easily interpretable by the eyeballing method. This aspect comes in favor of the agreement of Rosner et al. (2014) according to which the coherence score is a limited measure for accuracy and interpretability in the case of LDA. Thus, the next highest-scoring models were inspected.

As a result, the LDA model chosen to identify the topics of discussion of tweets was the one with 12 num topics, 0.03 alpha value, 150 iterations.

4.6.4. Sentiment Analysis on Tweets

Topic modeling the user descriptions delivered to the participants, i.e., the industries, of the scraped tweets. Next, the authors want to explore the perception of ChatGPT of the identified industries by conducting sentiment analysis on the enclosed tweets. As previously introduced, the preferred approach for polarity classification in this research is VADER, which is a lexicon rule-based approach specifically developed for social media analysis (Mandal et al., 2021). VADER, an open-source python tool, stands for Valence Aware Dictionary and Sentiment Reasoner. VADER offers the benefit of analyzing the sentiment of any text without the requirement for prior training of the underlying model and considering sensitivity to both polarity (positive/negative) and intensity (strength) of emotion (Calderon, 2017).

With VADER having capabilities to handle emojis, word capitalization, punctuation and more, little to no data pre-processing is required for the input data. Thus, VADER was used on the original tweets and not on the tokenized version of those. However, to remove the remaining noise URLs and mentions were completely removed. Additionally, a primitive function splitting the hashtags into individual words was created and used on the tweets.

Hence, the tweets stripped of URLs and mentions, were analyzed and classified with the following three categories based on the compound polarity score:

- Positive
- Negative
- Neutral

The compound score holds the overall sentiment polarity of a tweet and is the sum of positive, negative, and neutral scores which is then normalized between -1 (most negative) and +1 (most positive). A compound score between -0.05 and 0.05 was considered a neutral sentiment.

To assess the accuracy of VADER's outcomes the authors of this paper use a sample test set of 200 stochastic tweets. First, they were hand-labeled by the authors and then analyzed by VADER. Based on the differences between the two classifications the following accuracy measures resulted.

Case	F1 Score
Positive	0.63
Negative	0.38
Neutral	0.50
Compound	0.54

Table 6 - VADER Accuracy Scores on ChatGPT Tweets Dataset

4.7. Exploratory Data Analysis

The data analysis of the data with the aim of generating insights within the scope of this study was mainly conducted through calculations and visualizations in Microsoft's PowerBI.

Concomitantly, jupyter notebooks were created to support the enrichment of data quality or to support tasks where python scripts and visuals were considered more useful.

Among the tasks completed in python were:

- Creation of final analysis datasets
- Outlier removals
- Visualizations

As detailed in Appendix 6, before experimenting with data in PowerBI two datasets were created where all information considered relevant about tweets and users was recorded.

Additionally, two types of tweets were identified and considered redundant for the scope of this study as further exemplified in Appendix 6. These were categorized as such based on python scripts. Additionally, a new dimension about tweets has been added which highlights whether a tweet was created in response to another tweet or not.

The jupyter notebooks used in this part of the methodology were continuously expanded on a need-base with specific visuals and calculations.

While numerous fields were stored in the final data tables, as described in Appendix 6, each of the tables had certain fields which were primarily used in the analysis.

The fields used to analyze tweets are:

- tweet_id: stores unique identifiers of tweets
- sentiment: stores sentiment of tweet (e.g. Positive, Negative, Neutral)
- compound: stores compound score
- date: stores tweet date
- topic: stores tweet topic

- tweet_relevant: stores True if tweet is relevant and False if tweet is not relevant
- inreply_tweet: stores True if tweet was in reply to another and False if tweet was not a reply

The fields used in the analysis of users are:

- usr_userid: stores unique identifier of users
- first_tweet_date: stores date of first tweet created by user
- usr_industry_category: stores industry of each user

Based on the aforementioned data fields, various analysis methods were further employed in PowerBI and jupyter notebooks.

4.8. Summary of Methodology

Figure 8 depicts the course of actions in the methodology section. The authors started with scraping tweets and users from Twitter. Next, a first screening and cleaning of the data happened. After that, the pre-processing took place with removing noise, lemmatization and tokenization of the data so that the LDA modeling can work efficiently and effectively. By looking at different coherence scores for different number of topics, the authors determined that the most optimal number of topics to be eight. As two of the topics were unclassifiable, only the users belonging to the remaining six topics were further analyzed. Regardless of the fact that their scrutiny would bring knowledge about users with no particular industry, the authors believe that including them for further analysis and discussion will not add value to answering this study's research questions. Once the industries were defined through topic modeling, the tweet sentiment polarity was discovered with the lexicon-based VADER approach. In the end, the results of the sentiment polarity for all users, as well as the industry specific sentiment were analyzed through appropriate statistical methods and visualizations, created in PowerBI and jupyter notebooks. Finally, these were interpreted through the DOI theory to ascertain the acceptance of GLLMs and the state of diffusion.

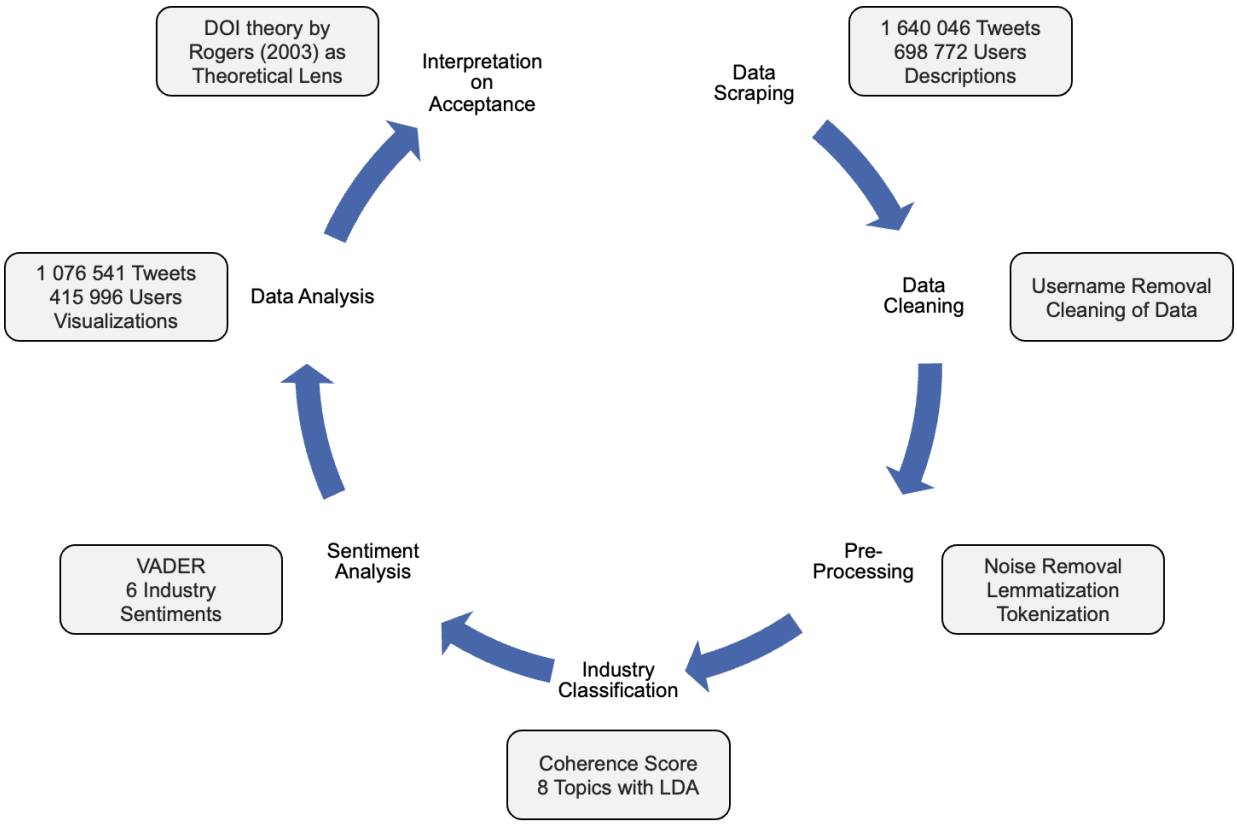

Figure 8 - Summary of Methods

4.9. Quality Evaluation

To assess the quality of the present research and its applied methods, validity and reliability are two key factors that must be taken into consideration.

4.9.1. Validity

The three main types of validity to be discussed are construct validity, internal validity and external validity (Saunders et al., 2019).

Construct validity refers to the extent to which a research study measures what it is intended to measure (Saunders et al., 2019). To ensure a high level of construct validity, the authors extracted and processed data with methods exclusively from trusted and peer reviewed sources that have aimed to deliver quality sentiment and topic modeling results. It was made sure that the data scraping was conducted with the appropriate search query and the collected data was cleansed extensively to ensure data validity. This would also lead to higher validity in the topic modeling. Additionally to avoid any falsities

in the research design, every step, including the code, was double-checked and precisely documented by the authors.

Internal validity aims to measure how accurate the findings under the chosen steps are (Saunders et al., 2019). As far as the authors verified, the scraped data showed no issues concerning missing or incomplete data that would affect the accuracy of the results. One point worth mentioning is that additionally to the data pre-processing steps, the authors assumed tweets that appeared in a repetitive and bot-like manner to be affecting the accuracy of the sentiment polarity. However, after removing the most obvious tweets of that manner, it turned out to be not that pivotal. Furthermore, the accuracy of the lexicon-based approach was measured against the manual annotation of sentiment of both authors with a sample set of 200 tweets. At this point, the authors acknowledge that the accuracy of VADER is not optimal, but it is still a widely accepted approach to automatically analyze the sentiment of a large data set. In terms of topic modeling, appropriate measures were taken to ensure that the resulting topics reflect meaningful industries.

External validity focuses on representativeness, which means to what extent findings can be applied beyond the specific case (Saunders et al., 2019). Considering that this paper aims to find Twitter user sentiment and user industries in order to explore the diffusion and acceptance of new innovations, the applied methods will generally not produce findings representative beyond this communication channel. This paper may not yield in generalizations beyond Twitter users, but it may portray valid insights in relation to the theory. However, the authors believe that the study provides a solid foundation of well-established methods that can be utilized to examine the industries and the related sentiments.

4.9.2. Reliability

There are several points related to the reliability of the findings of this paper. Reliability refers to the methods used in this study when repeated at a different time or by different researchers (Saunders et al., 2019). To guarantee reliability, the authors disclosed each part of this work in a fully transparent way to allow others to judge for themselves and to

replicate this study if they wished to do so. Except for the non-deterministic outcome of topic modeling, the results for data cleansing and sentiment analysis should provide qualities of trustworthiness and stability. To mitigate the threat for validity and reliability by arbitrarily classifying the topics found through topic modeling to industries, each specific topic was labelled by the two authors and any disagreements were resolved through discussion.

5. Findings

In this chapter, the findings resulted from analyzing the collected tweets with the help of LDA and VADER will be discussed with the scope of answering the Sub-RQs and the main research question of this paper: **How are generative large language models perceived by participants of different industries based on social media commentary?**

Moreover, the findings are structured based on the Sub-RQs.

Sub-RQ1: To which industries are users that take part in the social media commentary about ChatGPT belonging?

Based on the implementation of LDA it was observed that not all users participating in the conversations about ChatGPT on Twitter belong to an industry. However, from the sample of tweets considered relevant in this study (1 630 348), more than 59% (415 996) of the users could be mapped to an industry based on their user description.

These have been generated by a total number of 698 690 unique users with a ration of tweets-by-user of 2.3. However, 105 087 users had either no description or one solely consisting of special characters. Additionally, this resulted in 593 603 user descriptions being used in the LDA-based topic modeling process. The eight topics identified among these descriptions are: Software Engineering, Marketing & Business Development, Entrepreneurship & Business Leadership, Finance & Crypto, Education & Academia, and Arts & Entertainment. Additionally, two other topic clusters (T3, T8) were observed as shown in Figure 9. The keywords related to the last two topics do not present any relation to a particular industry.

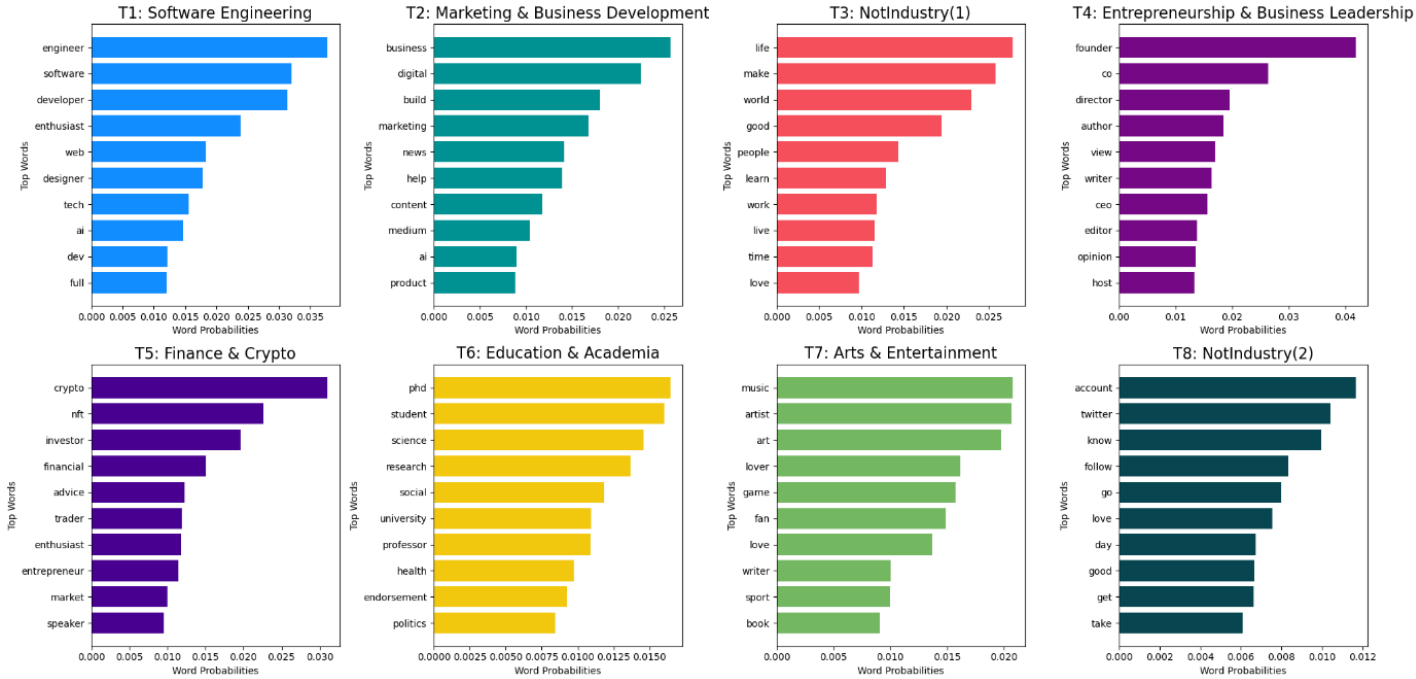

Figure 9 – Top 10 Words of LDA Topic Modeling on User Descriptions by Category

With the scope of this paper being grounded in the perception over GLLMs across industries, T3 and T8 will be excluded from the further analysis section of this paper, with rare exception where this will be specifically mentioned.

Thus, the 1 076 541 tweets generated by the 415 996 users belonging to an industry will make the object of this analysis. The overall ratio of tweet-per-user stands at 2.6.

The categorization into industries was done based on the top 10 words present in the collection of words making up each LDA output. Figure 9 shows the top 10 terms for each topic. Based on these terms, the authors used the visual inspection method to label each industry appropriately.

For T1, the tokens with the highest probabilities were “engineer”, “software” and “developer”. Hence, this topic was labelled as the “Software Engineering” industry. The count of users belonging to this industry is 89 028 and they have tweeted 218 381 times.

In the business segment, two industries could be identified. One topic (T2) contained mainly terms about “business”, “digital”, “marketing” or “content”, which the authors then labelled as the “Marketing and Business Development” industry. The count of users belonging to this industry is 93 372 and they have tweeted 353 077 times.

In line with the business aspect, the other topic (T4) revealed that the industry could be "Entrepreneurship and Business Leadership" through tokens such as "founder", "ceo", or "director". The count of users belonging to this industry is 52 422 and they have tweeted 122 573 times.

Next, T5 revealed with terms, such as "crypto", "investor", "financial" or "trader" that the respective industry would be "Finance and Crypto". The count of users belonging to this industry is 47 436 and they have tweeted 106 042 times.

Another clear topic for the authors is T6, which includes words such as "professor", "student", "science" or "university". The authors classified this topic as "Education & Academia"'. The count of users belonging to this industry is 65 601 and they have tweeted 148 372 times.

Furthermore, the next topic (T7) can be mapped as the "Arts and Entertainment" industry due to the top terms being "music", "artist" and "game". The count of users belonging to this industry is 68 137 and they have tweeted 128 096 times.

Figure 10 gives an overview of the distribution of users over the identified industries and the ratio of tweet count by industry users.

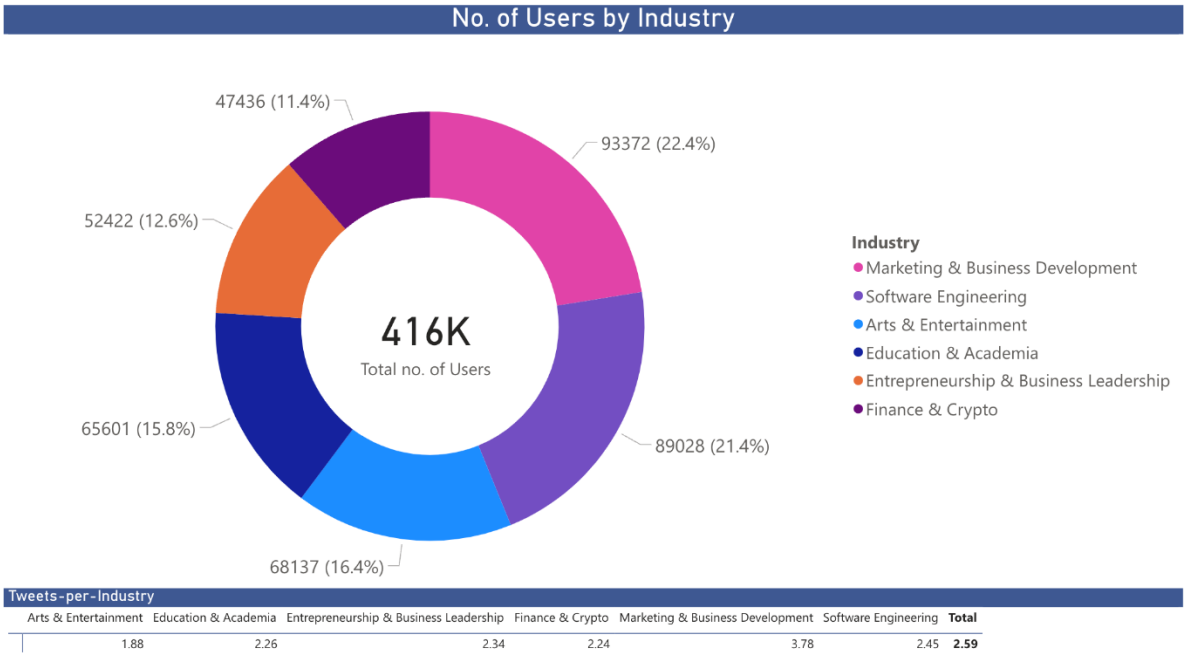

Figure 10 - Distribution of Twitter Users by Industry

To conclude, the industries showing interest in ChatGPT and thus in GLLMs are Marketing & Business, Software Engineering, Arts & Entertainment, Education & Academia, Entrepreneurship & Business Leadership, and Finance & Crypto.

Sub-RQ2: How is ChatGPT perceived across the identified industries?

When shifting the attention towards the sentiment observed among the total users of 415 996, it is noticeable that these recorded a 0.19 compound score as a result of the sentiment polarity classification conducted with VADER on their tweets.

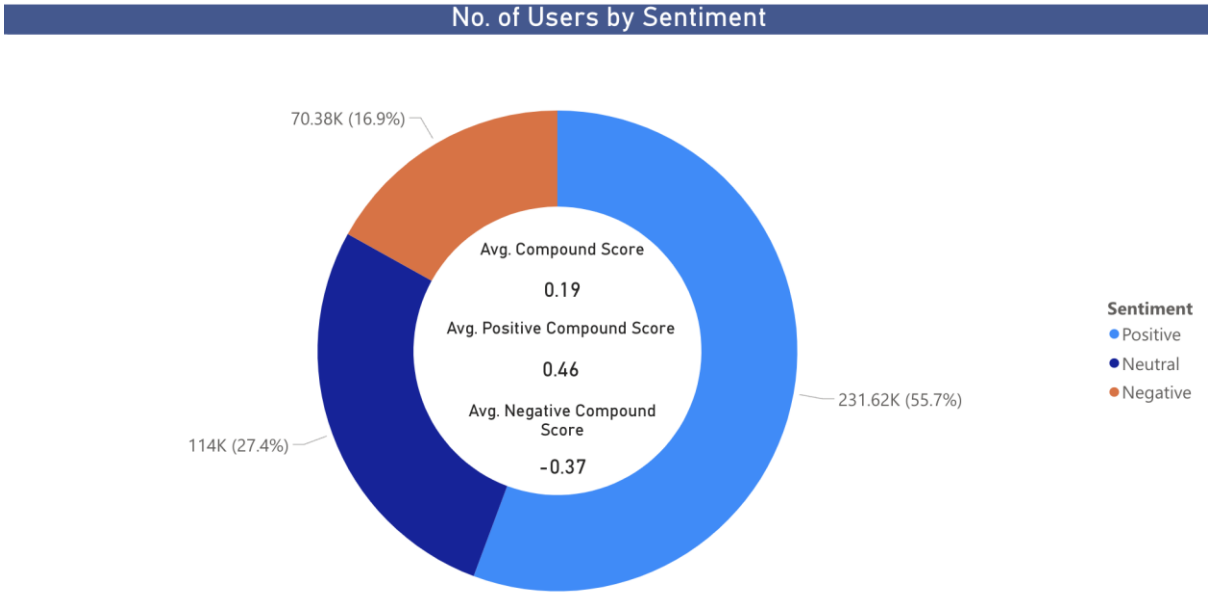

Figure 11 – Number of Users by Sentiment

As the overall polarity score is positive and higher than the threshold set in this project for the neutral category, it can be concluded that GLLMs are generally perceived in a positive manner. The average compound score recorded for positive tweets is 0.46 while, the one for negative tweets lies at -0.37.

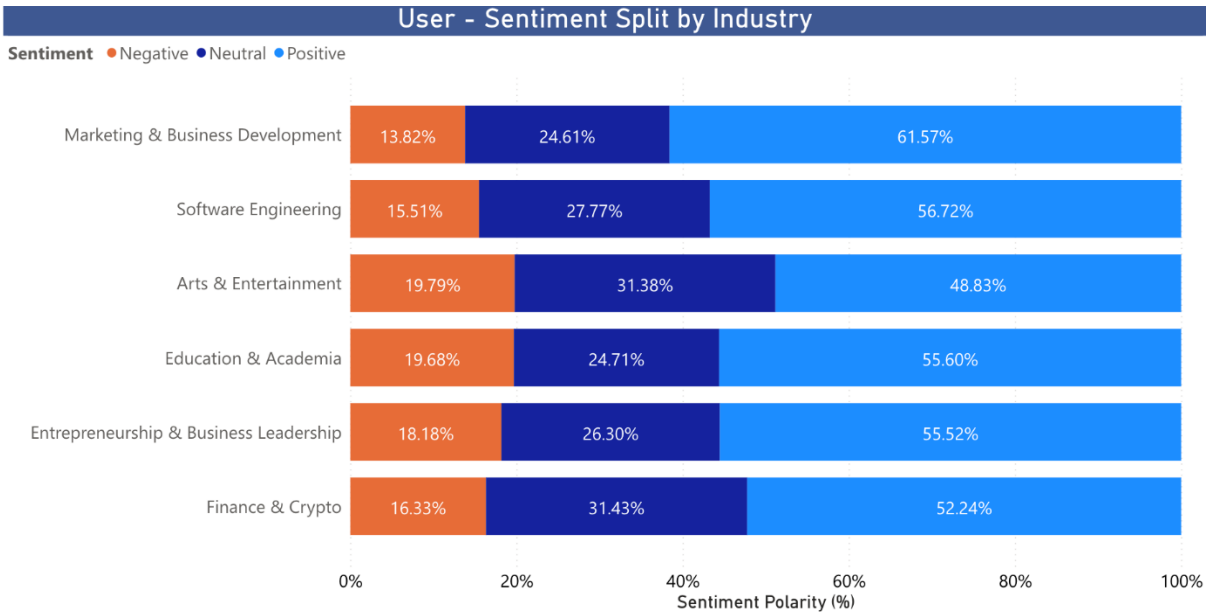

Figure 12 – User Sentiment by Industry

The industries that show the biggest portion of positive sentiment, i.e. acceptance, are the industries of "Marketing & Business Development", "Software Engineering" and "Education & Academia". These industries will eventually advance to implementation and finally, into the confirmation stage, that will disclose if ChatGPT will be in continuous use or not.

The biggest representers of neutral sentiment are industry "Finance & Crypto", "Arts & Entertainment". These industries have users with the tendency to have a bigger neutral stance. For individuals with neutral opinions, the authors suppose that their opinions will gradually change as the benefits and positive outcomes of the dissemination start to materialize.

By contrast, negative attitudes can be found more for the industries of "Education & Academia", "Arts & Entertainment" and "Entrepreneurship & Business Leadership".

The average compound scores for each industry are displayed in Table 7.

Industry	Average of Comp.	Standard Deviation of Comp.	Median of Comp.	Min Comp.	Max Comp.
Arts & Entertainment	0.15	0.39	0.000	- 1.00	0.95
Education & Academia	0.18	0.39	0.10	- 1.00	0.95

Entrepreneurship & Business Leadership	0.19	0.38	0.10	- 1.00	0.95
Finance & Crypto	0.18	0.37	0.05	- 0.98	0.95
Marketing & Business Development	0.23	0.36	0.20	- 0.98	0.95
Software Engineering	0.20	0.37	0.15	- 1.00	0.95
Total	**0.19**	**0.38**	**0.15**	**- 1.00**	**0.95**

Table 7 – Sentiment Polarity Statistics by Industry

When inspecting the distribution of values, it can be observed that moderate sentiments take a large part. As 27.4 % of users are holding a neutral sentiment, this impacts the distribution to revolve around 0. With that being said, the authors further investigated the sentiment statistics excluding the neutral segment to understand more about the distribution of the users that hold definitive perception.

Industry	Average of Comp.	Standard Deviation of Comp.	Median of Comp.
Arts & Entertainment	0.22	0.45	0.30
Education & Academia	0.24	0.43	0.30
Entrepreneurship & Business Leadership	0.25	0.42	0.30
Finance & Crypto	0.26	0.42	0.30
Marketing & Business Development	0.30	0.38	0.35
Software Engineering	0.28	0.41	0.35
Total	**0.26**	**0.42**	**0.30**

Table 8 – Polarity Statistics for Negative and Positive Perceptions by Industry

It can be observed that the most homogeneous perceptions across users are in Marketing & Business Development with a standard deviation of 0.30, while the Arts & Entertainment industry have the most parsed opinions with the standard deviation of 0.45.

To understand how conversations about ChatGPT have developed since its launch, the overall sentiment of tweets was analyzed from a longitudinal perspective.

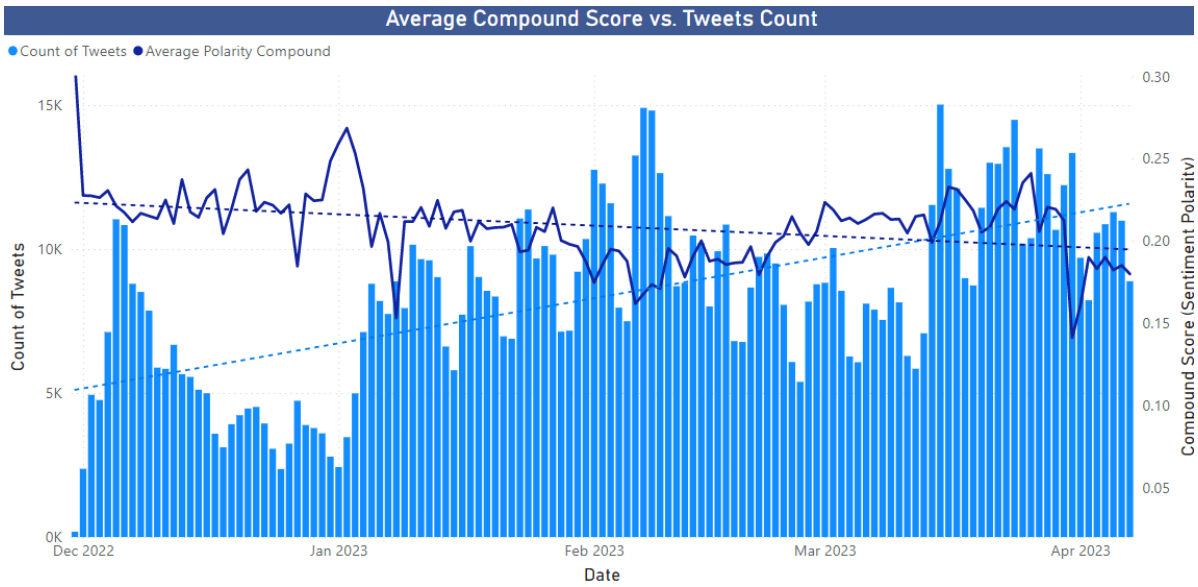

Figure 13 - Average Compound Score vs. Tweets Count

Figure 13 shows that the most amount of tweets were recorded in March 2023, peaking on the 15th of March at 15 020 tweets generated by 11 700 Twitter users. The average compound score peaked on the day of ChatGPTs release. However, this was expected since only 186 tweets about the product were recorded, with a vast majority of those announcing the product's launch. Thus, other movements in the average compound score are considered to be more representative. While the score has been positive overall, fluctuating between 0.14 and 0.31, the average compound score across all industries follows a trend of slight decline. Certain peaks and considerable drops can be observed in Figure 13.

On January 2nd of 2023, the highest peak of the average compound rate, excluding the date of launch, was recorded at 0.27, proceeded by a second high with an average compound score of 0.24 recorded on the 26th of March. The most significant drops in sentiment were recorded on the 8th of January, 6th of February and 31st of March, with values of 0.15, 0.16 and 0.14 respectively.

As opposed to the decreasing trend of the average compound, the number of tweets show an increase in volume.

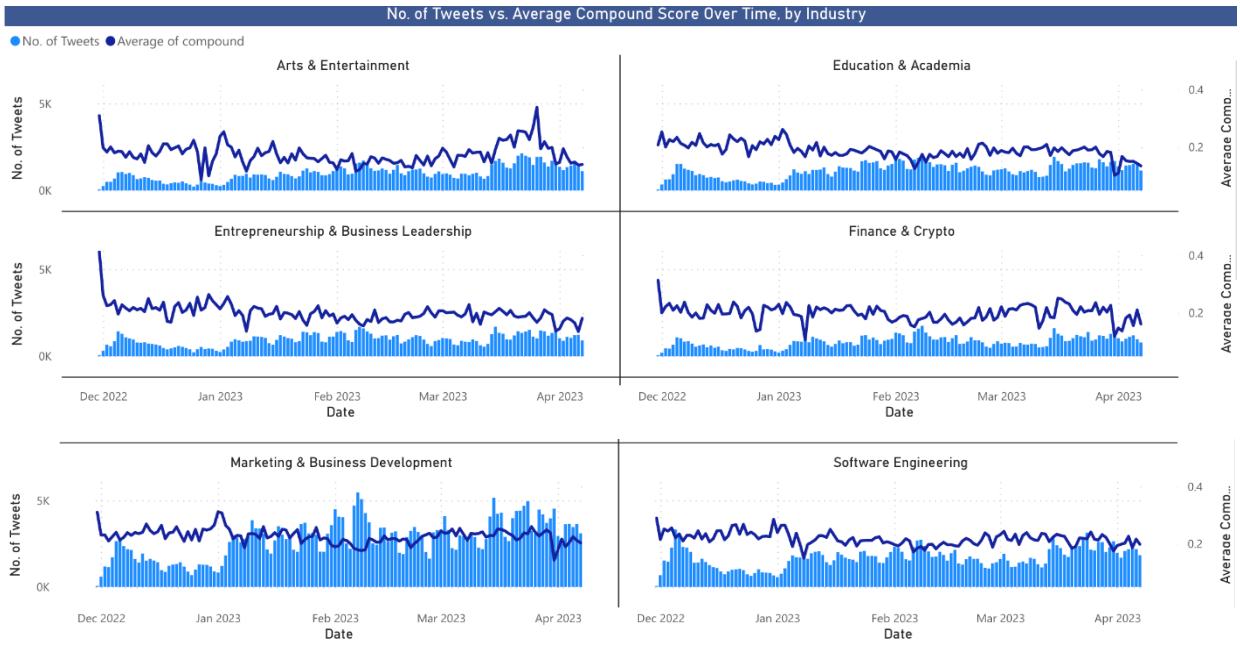

Figure 14 - Average Compound Score vs. Tweets Count by Industry

At an industry level, it can be noticed that all industries experience a decrease in sentiment in the first days of April, as documented in the PowerBI dashboard "Average Compound Score vs. Tweets Count" in Appendix 8. Additionally, the data analysis showcases the peak in the number of tweets and the drop in sentiment recorded on 8th of January for all industries. The most significant drops were recorded for the Arts & Entertainment space (0.21 to 0.09), and Software Engineering (0.24 to 0.12). In terms of volumes, the two industries identified present interesting particularities. While the volume of tweets about ChatGPT started high for Software Engineering, the starting volume for Arts & Entertainment was shy. The first shows a decrease during the later part of January, and the latter an increase.

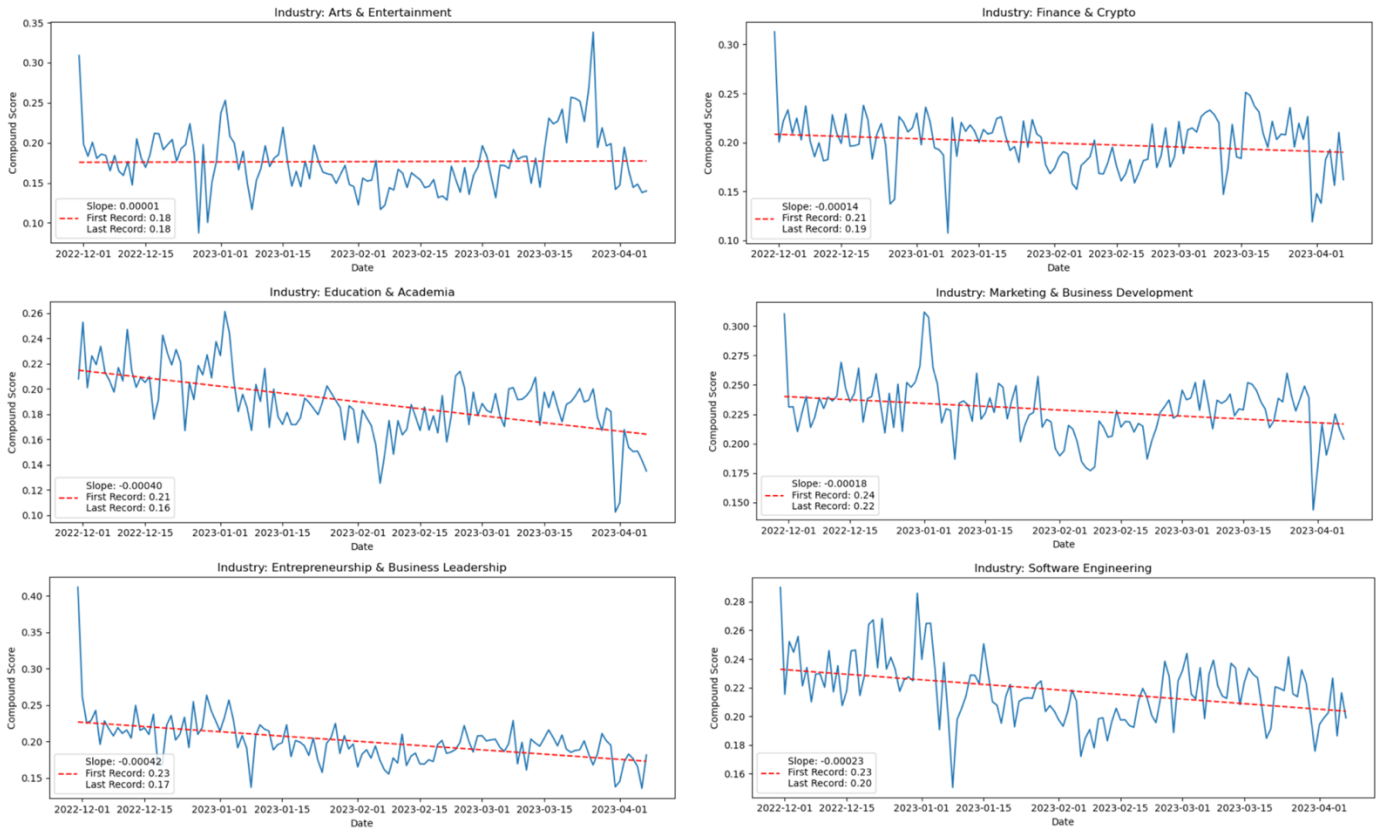

Figure 15 - Average Compound Score by Industry Over Time

In terms of the overall change in sentiment over time, Entrepreneurship & Business Leadership, Education & Academia and Software Engineering are displaying the highest dropping rates. As further detailed in the "Sentiment trend on Tweets" section of Appendix 6, the trend of the compound rate for Entrepreneurship & Business Leadership follows a decline of -0.00042, while the trends for Education & Academia and Software Engineering show decreasing rates of -0.00040 and -0.00023 respectively. Across all other industries, a negative relationship between the passing of time and the sentiment over ChatGPT can be observed, except for Arts & Entertainment. For the latter industry, the trend shows a stable perception over time.

As an additional layer to the research, all tweet contents were further investigated and categorized into topics, too. As described in Section 4.5.3., LDA was applied in identifying the most commonly discussed topics in the tweets related to ChatGPT. The topics resulting from LDA are: Learning & Education, Human-Bot Interaction, AI R&D Competition, Future of Jobs & Societal Impact, Content Generation, ChatGPT Features & Accessibility, two topics related to Problem Solving and two topics which could not be

mapped to a clearly understandable topic. For further analysis, the two Problem Solving topics were viewed together as one topic. The naming convention for the topics generated by the LDA model, has been attributed based on the top 10 words belonging to each topic depicted in Figure 16.

Figure 16 – Top 10 Words of LDA Topic Modeling on Tweets by Category

The category discussed the most across all tweets in this study is the Problem Solving one, counting no less than 294 188 tweets with an average positive compound score of 0.23. When comparing compound scores, however, Problem Solving is lagging behind Content Generation which scores a 0.28 compound score. The topics with the lowest compound scores are Future of Jobs & Societal Impact (0.12) and Learning & Education (0.09).

Sub-RQ3: What do tweets about ChatGPT indicate about the acceptance of GLLMs?

The DOI theory places importance on a social system in the process of traction and diffusion of an innovation that results in the acceptance of a new concept, or technology.

The data collected with the scope of the present research refers to opinion sharing in a public forum (Twitter) about ChatGPT, which is used as a proxy for GLLMs. Thus, the results related to ChatGPT are interpreted as representative for GLLMs across this paper.

Consequently, with DOI underlining the dependency of adoption on perception is used to interpret the findings of this study.

With 27.4% of the Twitter pertaining to an industry holding a neutral sentiment, it can be argued that the authors of these tweets are in the knowledge phase of the decision-making process. This implies that they are aware of the product but have yet to form an opinion regarding its acceptance.

This project assumes that the users that have formed an opinion have overcome the knowledge phase and are now in the persuasion stage. With 55.7 % tweets being positive it can be assumed that their authors will lean towards the decision to adopt the technology and continue to make use of it.

The negative segment discloses that 16.9% of users are likely to decide that this technology shall not be adopted at all or that they will lag in adoption.

At an industry level it can be observed that the majority of users pertaining to each industry have a positive perception over ChatGPT, thus GLLMs. With the exception of Arts & Entertaining, the overall tendency towards adoption is reflected in each industry. Moreover, Marketing & Business Development or Software Engineering, have concomitantly recorded the largest proportions of users having a positive sentiment, and the least number of users having a negative sentiment. These two industries are thus, displaying the highest tendencies for adoption.

The large proportions of users, that belong to the industries of Finance & Crypto and Arts & Entertainment, holding a neutral perception highlight that the participants of these industries may find it harder to overcome the Knowledge phase of DOI. In addition, with the least amount of users being categorized as neutral, Education & Academia is the industry displaying the most split acceptance behaviors.

On another note, Arts & Entertainment is recording one of the highest numbers of users holding a negative attitude. The second highest proportion of users showcasing negative attitudes is recorded by Education & Academia. Thus, the participants of the two industries show the most concern and reluctancy in adopting the technology.

6. Discussion

This chapter entails the key points worth discussing according to the authors. It is structured to critically review the findings to answer the research questions. Furthermore, the outcomes are assessed with evidence found from peer researchers. This leads the reader to the implications and potential generalizations that can be generated through examining Twitter perceptions on GLLMs. Lastly, the applied methodological and theoretical strengths and limitations of this study are gaged.

Consequently, this chapter follows four focus topics for the discussion:

1. Summary of Main Findings
2. Implications
3. Strengths and Limitations

6.1. Summary of Main Findings

The present paper aimed to answer the research question: **"How are generative large language models perceived by participants of different industries based on social media commentary?"** To do so, three supporting sub-research questions were posed.

The first one aims to uncover which industries are represented in the content generated on Twitter about ChatGPT. With that being said, the authors chose to study Twitter user descriptions and to identify whether these contain relevant information about the users' domains of professional activity. To be able to conduct this task, LDA was deployed. The use of LDA comes in alignment with the studies from Haque et al. (2022) and Taecharungroj (2023) that have also used the algorithm in topic modeling with the scope of exploring. As a result of the topic modeling process, it was identified that users from the following industries are participating in the Twitter discussions about ChatGPT: Marketing & Business Development, Software Engineering, Arts & Entertainment, Education & Academia, Entrepreneurship & Business Leadership, Finance & Crypto.

The results are considered satisfactory due to multiple reasons. In the process of using LDA, the authors have tuned the parameters of the algorithm with the aim of achieving

meaningful topics. With that in mind, the LDA model used had a coherence score of 0.56, which is considered a good score by the data science community, and generated topics which are reasonable for human classification. An additional point that legitimizes the results for Sub-RQ1 is that the studies that come closest to this research, namely from Haque et al. (2022) and Taecharungroj et al. (2023), have also employed LDA. The difference is that the two other studies have applied LDA solely on Tweet contents. Nonetheless, Haque et al. (2022) disclosed occupations of the authors of their examined tweets. Occupations such as “student” and “academics/researcher” that would fall into this study’s Education & Academia industry. Other occupations that correspond to the identified industries would be “software practitioner” that matches the Software Engineering industry or “journalist” and “entertainer” that fall into the industry of Entertainment & Media. Altogether, findings for Sub-RQ1 are congruent with studies that focus on the area of perceptions on GLLMs.

For Sub-RQ2, the present paper aims to understand the perception of GLLMs among the users that are part of an industry. The findings lead to the conclusion that the overall perception is positive across users from all industries, with the positive sentiment prevailing the neutral and negative sentiments. It can be claimed that this outcome is in line with the previous findings from the existing literature. Taecharungroj et al. (2023) concluded that ChatGPT has mostly received positive reception on Twitter after the first week of launch. Additionally, Haque et al. (2022) deduce that their sample data set gave them results with overwhelmingly positive sentiment, especially in the software engineering and entertainment topics. This study’s findings show that the overall sentiment, as well as industry specific sentiments, such as in the industry of Software Engineering, are still conforming previous findings about GLLMs after several months of usage and discussions.

To learn more about the factors impacting the sentiments, this study has provided a longitudinal study on how tweets about GLLMs have developed from the 30th of November 2022 to the 7th of Apr 2023. Through the longitudinal perspective, trends in tweets and sentiments with their most significant drops and peaks were also analyzed.

The authors contend that the present perceptions seem to be impacted by external factors. External factors that could cause volatility in the tweet sentiment polarity are significant social, economic and political news (Ahmad et al., 2021), as well as product and service announcements (Malhotra & Malhotra, 2015), which may evoke strong emotional reactions on Twitter. The findings of this study suggest that some key announcements and events may have had a considerable effect on the sentiment. A timeline of major events around ChatGPT is summarized by Stringer & Wiggers (2023).

One of the events happened on the 8th of January 2023, when the first news about ChatGPT being banned from New York schools emerged (Roose, 2023). This date coincided with the first significant drop in overall sentiment. In general, when looking into the topics discussed in the tweets, it is noticeable that Education is the topic consistently generating the lowest compound scores (see PowerBI Dashboard "Tweet Content Topics' Sentiment".

The second significant drop has been recorded on the 6th of February 2023, which coincides with the date when Bard, Google's GLLM was announced (Sparkes, 2023). A factual error made by Bard in an advertisement raised fears that GLLMs pose the risk of creating misinformation on a mass scale and that they are not ready to be integrated into search engines (Sparkes, 2023).

The last example for a significant drop would be the 30th of March 2023, when Italy's data protection authority temporarily banned ChatGPT over the breach of the EU's General Data Protection Regulation (GDPR) (McCallum, 2023). Concerns and lack of transparency about how OpenAI uses data that come through user input as well as data for training ChatGPT made the company restrict assess in Italy. Another reason is the fact that ChatGPT is known to confidently create inaccurate information on named individuals. This appears to be a concern in the EU because the GDPR gives people a number of rights over their information, including the right to have inaccurate information corrected and the right to be forgotten. It is unclear at this time whether OpenAI has a system in place to comply with these rights. When looking at the reactions on Twitter for the 30th of March 2023, it dropped to the lowest compound polarity score of 0.14, suggesting this adverse event contributed to a slip in the perception and hence, acceptance.

Although there is no significant change in sentiment noticeable for other events, the number of tweets spiked when OpenAI announced on the 1st of February 2023 that ChatGPT Plus, a paid subscription service, would launch. In line with this announcement, the reports about Microsoft launching Bing with built-in ChatGPT on the 7th of February and the launch of an ChatGPT Application Programming Interface (API) on the 1st of March, caused a similar high in tweet numbers.

To sum it up, the findings and discussions are considered sufficient for answering the Sub-RQ2. GLLMs are perceived in a positive light across industries. However, it should be noted that the sentiment has decreasing tendencies for all industries apart from Arts & Entertainment. Another point worth mentioning is that external factors such as the reluctancy from the educational sector and pushbacks of governments seem to impact the perceptions negatively.

The third and last research question tries to answer what the perceptions over ChatGPT indicate about the potential acceptance and diffusion of GLLMs. While Haque et al. (2022) touches upon future acceptance of ChatGPT in their implications, Taecharungroj et al. (2023) conclude with ChatGPT's impacts on technology and humans. But both lack to induct their findings back to any technology acceptance theory.

This study, however, intends to draw inferences on diffusion and technology acceptance with the DOI theory by Rogers (2003). Based on the findings in Sub-RQ1 and Sub-RQ2, it can be concluded that the possibility of diffusion and acceptance is high due to the overall positive sentiment.

Additionally, it was observed that the number of users joining the conversations of ChatGPT on Twitter is still growing. These are continuously aligned with the overall sentiment proportions. As such, it can be assumed that the GLLM technology is still in the early phases of diffusion. This can also be underlined by the fact that, while ChatGPT was launched in Nov 2022, it was not until March when a peak in the number of tweets about the topic was reached.

In this context, when looking at the adoption curve proposed by Rogers (2003), the users that are part of this study can be categorized as innovators, since they are part of the first 2.5% of Twitter users discussing this novel technology.

It is, however, worth mentioning that according to Hagger (2019) theories including the DOI model can only provide partial explanations of behavior. Instead, these provide a framework for explainability of a phenomena and potential assumptions about the adoption of a technology.

To conclude, GLLMs are benefiting from an overall positive perception across the industries identified in this study. The longitudinal analysis of the sentiments revealed in tweets has further underlined this as there was little tendency towards sentiments becoming negative. Analyzing Twitter data has uncovered that Marketing & Business Development together with Software Engineering are the industries that show the most activity and the most willingness to adopt the GLLM technology.

6.2. Implications

The implications of the present study can be divided into two perspectives: a theoretical one and a practical one.

When discussing the theoretical perspective, the study showcases the applicability of the DOI model to the social network environment with the purpose of identifying trends related to the adoption of a technology. This highlights the role of user descriptions in mapping which industries are most present in the social media discussion about a technology, which in this case refers to GLLMs. Together with mapping the sentiment polarity of the tweets generated by the members of an industry, assumptions about the diffusion process of GLLMs can be made. Additionally, the study's longitudinal nature provides further insight on whether the sentiments generated about a technology experience high fluctuations over time, or whether they display a steady perception among the industries discussing the technology.

From a practical perspective, the present study proposes the use of ML algorithms for extracting topics from users' descriptions to identify industry participants that show interest in a technology. The further employment of sentiment polarity tools such as VADER to identify the overall perception over a technology, underlines its usefulness in assessing the potential of diffusion of technologies.

With a starting point in the theoretical and practical implications of this study, its generalization can be discussed related to two perspectives. Firstly, the choice of technical methods and their applications on the tweet's dataset provides a framework which could be further applied in the context of other technologies. While LDA and sentiment analysis have previously been employed in academic research, the application of the topic modeling techniques to user descriptions with the scope of identifying industry cluster, has yet to be popularized, with few prior studies such as Haque et al. (2022) placing importance to user descriptions in identifying professions.

Secondly, by incorporating a social science theory together with ML techniques, the present paper gains the characteristics of a CSS study that has the scope of exploring opinions shared on a social network to draw conclusions about the phenomena of technology adoption.

From a practical perspective, the overall positive sentiment over ChatGPT and thus, GLLMs, could serve as an indicator about the overall tendency towards acceptance and adoption across different domains ranging from Software Engineering to Education and to Business related fields.

In this context, with topics such as Problem Solving and Content Generation leading in terms of sentiment, we recommend for representatives of every industry to assess diligently the potential of the technology based on data privacy, cyber security, and job adaptability considerations.

Finally, the adoption of GLLMs should be a focal point of discussion for legislators, as currently large corporations are driving GLLMs development, which may lead to a disparaging increase in power over consumers.

6.3. Strengths and Limitations

Bearing the difficulties of studying social media data in the context of sociology theory in mind, this section intends to address the strengths and limitations of this study's particular approach. This reflection stems from the responsibility as researchers to consider the theories employed, and their implications. Particularly for pragmatism research, it is important to reflect on areas of improvement and to objectively assess the methods used to guide further research and potentially, provide scholars with a foundation for prospective endeavors.

6.3.1. Strengths

The first strength of this study is the time and scope of the sample data. The longitudinal character and the extensive scope, that comprises every tweet in English containing ChatGPT in the defined timeframe, allowed the authors to examine the development of an innovation's sentiment with a representative sample of overall tweets. The combination of commonly accepted methods to collect, clean and process Twitter data provides future researchers a repository of tools that can be implemented quickly, effectively and reliably. The authors have been mindful of ethical considerations through anonymization for the data that has been processed and the outcomes that have been produced.

Furthermore, this study's extensive nature of using the DOI theory for sentiment polarity interpretation can be seen as a strength. Limited research has been conducted that follows the principles of this social theory in a close manner. The authors computed the users analyzed into the exact category of adopters of the adoption curve and provided the rationales to map positive, neutral and negative sentiments to the decision-making process stages. This theoretical lens enhanced the ability to identify patterns that may have not been visible if the DOI theory was excluded.

6.3.2. Limitations

This study demonstrates how Twitter may be utilized to gather data sets that can shed light on fast emerging technologies like GLLMs. There are, however, a number of drawbacks to this research that must be acknowledged.

To begin with, the authors want to highlight that conducting the literature review posed some difficulties as new research related to ChatGPT and GLLMs appeared almost every week. Hence, relevant papers that could have been addressed and taken into consideration for the literature review, as well as for the discussion section might have been missed.

Secondly, the authors examined Twitter as a digital communication channel and the only data source due to its influential nature and the presence of industry opinion leaders. Alternative digital communication channels, such as Reddit, Facebook, Instagram, and YouTube, could all provide large datasets related to discussions about ChatGPT. However, due to time constraints these have not been explored. Potentially, these other platforms may paint a different picture of the diffusion of GLLMs.

Moreover, Twitter is available globally and communication happens in multiple languages but only tweets written the English language were considered for this study, which means that there is a linguistic limitation. In the strengths, the authors mention that the data sample might be considered as a representative sample of overall tweets about ChatGPT in English. Thus, this study should not be considered a representative global Twitter sample. Besides, this study does not center on demographics data of the users. Sentiment and diffusion might look different among distinct age groups, genders and more.

In addition to this, existing comparable systems to ChatGPT, such as Bard from Google, NeMo from Nvidia or LLaMA from Meta have been taken out of scope for this research and such keywords were not included in the data scraping process. Their exclusion can be explained by the fact that they have not been widely available for end users yet.

However, this implies that the diffusion of GLLMs may be different once such products become widely available.

In addition, to address the study topic, a variety of different statistical and natural language processing tools might have been applied to the dataset. However, due to computational and time limitations, an appropriate selection had to be made. The use of alternative methods such as BertTweet for sentiment analysis, or BertTopic for topic modeling could have provided a space for comparison between models. This could have led to even more cohesive results both in terms of sentiment polarity, and topic modeling. However, Haque et al. (2022) and Taecharungroj (2023), that served as reference and guidance studies used LDA. Moreover, the results of LDA are also not deterministic, signifying that every iteration of LDA would have brought a different outcome than the one presented in the findings.

In terms of sentiment polarity, the authors found that VADER underperforms on this study's data set in comparison to the accuracy scores attained by Hutto & Gilbert (2014) on the datasets used in their paper. When the authors examined tweets with the utmost polarity in detail, an overuse of exclamation marks and emojis loomed. Furthermore, VADER is not capable of capturing jargon, including the meaning of emojis in different contexts and misspellings, which are common on social media networks. The authors presume that all these add up to an acceptable, yet not excellent, F1 accuracy score of 0.54.

There are several weaknesses of the DOI theory that concern this study. First, the stage model cannot display the complete innovation cycle. For instance, during the diffusion phase, innovation may change significantly. It assumes that an innovation is a fixed, stable entity that remains unchanged over time, but in reality, innovations can evolve and change in response to user feedback and societal needs. This actually happened, for instance, when ChatGPT Plus with the updated GPT-4 model launched. Furthermore, using Twitter sentiment data limited the authors to make solid assertions around the five perceived characteristics of innovations and the implementation and confirmation stages of the decision-making process. The authors claim that the emphasis lies on individual

actors, while the larger social, cultural, and economic factors that may affect an innovation's adoption are given less significance. Despite the fact that this theory has a focus on individuals, it is not considering a person's resources or the reach of their social network during the adoption process. Networks on Twitter are defined on the base of a follower system and the magnitude of a user's network determines the spread of information (Kiss & Buzás, 2015). Neglecting all this could lead to an oversimplified understanding of how inventions spread throughout society.

In addition, the authors assumed that sentiments express the best fit in the knowledge and persuasion stage of the decision-making process. It has to be acknowledged that this is not a sound concept. Tweets about ChatGPT do not prove if user have accepted it in practice. Furthermore, after the persuasion stage, people are in a continuous evaluation phase. While people are in the implementation and confirmation stage, it is presumable that they are also tweeting about their experience. This also means that the authors cannot state if the sentiment analyzed is from people who have already used GLLMs for a longer time, who, thus rather belong to the confirmation stage than the persuasion stage.

7. Conclusion

This study explored the sentiment of Twitter users towards ChatGPT and its level of acceptance as a new technology. Furthermore, the industries to which these users belong were identified, to understand what type of sentiments are held among these industries in relation to GLLMs.

The findings suggest that ChatGPT has received favorable responses overall, with the majority of tweets bearing positive sentiment about this innovation. The study also revealed which industries are more concerned and less acceptive. It is highly likely that the underlying technology of GLLMs will continue to gain acceptance and popularity in the future as this innovation is still at the early beginning of the adoption curve. Nevertheless, it will stay relevant for researchers and businesses to monitor how the overall perception and the stances of different professions will change in the future as it

is claimed that this technology has the potential to affect a quarter of all jobs (Vallance, 2023).

This research contributes to understanding the public perception of the GLLM ChatGPT and expands the literature of technology acceptance and computational social science. To conduce to this understanding, more than 1.6 million tweets that span over 4 months were extracted initially. For reference, the sentiment analysis by Haque et al. (2022) only had around 18 thousand tweets collected over two days. Still, this research's results are in line with those of Taecharungroj (2023) and Haque et al. (2022), who studied early reactions about ChatGPT. All three studies indicate that generative AI would gradually get accepted. Unlike the other two studies, this research focuses on Twitter user descriptions to make assertions about the industries of the users that participate in the discussions around ChatGPT. This was achieved through LDA topic modeling. The results of this were six distinct industries which allow researchers to gain comprehension about under-represented groups and their opinions and concerns. Furthermore, the findings point out the industries that host more innovators and industries that might be lagging in adoption. Additionally, this study also presents a longitudinal perspective, which was not considered by the other studies.

Taking everything into account, it became evident that the positive polarization of ChatGPT consistently outweighed the negative feelings about it. Consequently, with this result in mind, the DOI theory explains how the overwhelming favorable attitudes about ChatGPT have led to its spread on Twitter and beyond in record time. By combining the findings with Rogers' (2003) innovation–decision process, this study also demonstrates the novelty for the utility of social theory in social media analysis.

With the real-world benefits of the innovation demonstrated so far, future acceptance seems auspicious. Hype usually moves in and out of fashion and it is hard to predict what happens after the initial peak of expectations. However, this may not be the case for this technology, as people across industries are adopting GLLMs quickly due to their public availability and general-purpose application. There are no significant signs of decreased enthusiasm over the past few months as people acknowledge the viability and the potential applications that seem to be multifarious. It can be assumed that a sharp rise

in adoption begins if creators and regulators find the right balance for limiting the harm and risks that come with it. It is beyond doubt that GLLMs are here to stay. As it stands, the novelty factor and honeymoon phase are still persisting. Sooner or later, the majority of people will take this new level of natural conversations with machines for granted and the authors expect GLLMs to transform businesses and people's lives quickly once the magnitude of its capabilities is fully understood. By providing insights into the acceptance of GLLMs, this research may have significant implications for researchers, businesses across industries, policymakers, and society. The authors hope to impact especially researchers and creators of this innovation to seriously assess the negative sentiments overall and by industries that may be grounded in critical concerns. Players that fight for market share might move too fast and ignore safety measures that could lead to pitfalls only perceptible in a few years. To avoid that, a development with foresight and considerations on ethical, privacy and security aspects for everyone should be ascertained.

7.1. Future Research

Currently, a lot of research around GLLMs is happening and the authors want to highlight several key areas that need further research in this section. Firstly, this is a contemporary study that considers Twitter data to explore industry perceptions about ChatGPT. As mentioned before, whilst the rate of adoption is extremely high (Hu, 2023), mass adoption is still in its early phases. Researchers could repeat this study after some time has passed and expand the scope in terms of the timeframe to follow further developments in sentiment and diffusion of GLLMs. Alternatively, similar studies could be conducted on specific geographical regions or other languages. Different communication channels and social media networks such as Reddit, Instagram or YouTube should also be scrutinized as they offer valuable and rich discussions around GLLMs that would uncover more sentiments. This would eventually allow to spot differences between social networks and their role for the diffusion of technologies. Additionally, the GLLMs of OpenAI's competitors, such as Google, Meta or Baidu could be taken as the object of further research. Consequently, these potential studies could

complement or serve as comparison to the findings found in this study to push generalization, and to get a bigger picture on the diffusion of GLLMs.

Another path worth investigating is to supplement this study's findings with a more qualitative approach. By looking at tweet contents or conducting interviews with Twitter users that participated in the discussion about ChatGPT and GLLMs, future researchers would be able to not only understand the sentiment metrics but also the underlying messages. Furthermore, qualitative research would allow investigating the factors that influence the adoption such as the five perceived attributes of an innovation in the DOI theory or user demographics, that would allow the employment of other technology acceptance models such as the TAM or UTAUT.

Regarding methods used for topic modeling and Twitter sentiment analysis, the observed weaknesses around VADER could be addressed in further research. It can be contemplated to use ChatGPT or other GLLMs to conduct sentiment analysis and eventually, compare its performance with the ones that are currently used such as VADER or BERTweet.

Notwithstanding the fact that diffusion and technology acceptance are relevant, future research could also scrutinize the impact of this new innovation on established (digital) business models, labor markets and highly affected industries such as software engineering or content creation. Lastly, the authors advocate researchers to feel the duty for future research on GLLMs and their long-term implications on society and the environment as they are immeasurable and simultaneously inevitable to ensure a safe integration of GLLMs into society.

8. References

Acumen and Research Consulting (2023). Generative AI Market Size – Global Industry, Share, Analysis, Trends and Forecast 2022 – 2030. Retrieved 2nd February 2023 from: https://www.acumenresearchandconsulting.com/generative-ai-market

Adwan, O. Y., Al-Tawil, M., Huneiti, A. M., Shahin, R. A., Zayed, A. A. & Al-Dibsi, R. H. (2020). Twitter Sentiment Analysis Approaches: A Survey. International Journal of Emerging Technologies.

Ahmad, W., Wang, B. & Xu, H. (2021). Topics, Sentiments, and Emotions Triggered by COVID-19-Related Tweets from IRAN and Turkey Official News Agencies. SN COMPUT. SCI. 2, 394. https://doi.org/10.1007/s42979-021-00789-0.

Ajzen, I. (1985). From intentions to actions: A theory of planned behavior. Journal of Applied Social Psychology, 50(2), 179-211.

Al-Razgan, M., Alrowily, A., Al-Matham, R. N., Alghamdi, K. M., Shaabi, M. & Alssum, L. (2021). Using diffusion of innovation theory and sentiment analysis to analyze attitudes toward driving adoption by Saudi women. Technology in Society, 65, 101558.

Alsaeedi, A. & Khan, M. Z. (2019). A Study on Sentiment Analysis Techniques of Twitter Data. International Journal of Advanced Computer Science and Applications.

Balcioglu, Y. S. & Çerasi, C. Ç. (2023). Sentiment Analysis on YouTube: For ChatGPT.

Bhattacharya, P., Zafar, M. B., Ganguly, N., Ghosh, S. & Gummadi, K. P. (2014). Inferring User Interests in the Twitter Social Network.

Bian, J., Yoshigoe, K., Hicks, A., Yuan, J., He, Z. & Xie, M. (2016). Mining Twitter to Assess the Public Perception of the "Internet of Things".

Bickart, B. & Schindler, R. M. (2001). Internet forums as influential sources of consumer information. Journal of interactive marketing, vol. 15, 3. Pp. 31-40.

Black, I. (2006). The presentation of interpretivist research. Qualitative Market Research: An International Journal, 9(4), 319–324.

Blei, D. M., Ng, A. Y. & Jordan, M. I. (2003). Latent Dirichlet allocation. Journal of Machine Learning Research, 3, 993-1022. Retrieved 12th April 2023 from https://www.jmlr.org/papers/volume3/blei03a/blei03a.pdf

Bonta, V. & Janardhan, N. K. N. (2019). A comprehensive study on lexicon based approaches for sentiment analysis. Asian Journal of Computer Science and Technology, 8(S2), 1-6.

Borghi, M. & Mariani, M. M. (2022). The role of emotions in the consumer meaning-making of interactions with social robots. Technological Forecasting and Social Change.

Bouazizi, M. & Ohtsuki, T. (2015). Opinion Mining in Twitter: How to Make Use of Sarcasm to Enhance Sentiment Analysis. International Conference on Advances in Social Networks Analysis and Mining.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A. & Agarwal, S. (2020). Language models are few-shot learners. Advances in neural information processing systems. 33:1877–901.

Caldarini, G., Jaf, S. & McGarry, K. (2022). A Literature Survey of Recent Advances in Chatbots.

Calderon, P. (2017). VADER Sentiment Analysis Explained. Medium. Retrieved 4th January 2023 from: https://medium.com/@piocalderon/vader-sentiment-analysis-explained-f1c4f9101cd9

Cardona, D. R., Werth, O., Schönborn, S. & Breitner, M. H. (2019). A Mixed Methods Analysis of the Adoption and Diffusion of Chatbot Technology in the German Insurance Sector. Proceedings of the 25th Americas Conference on Information Systems (AMCIS).

Carvalho, J. & Plastino, A. (2021). On the evaluation and combination of stateoftheart features in Twitter sentiment analysis. Artificial Intelligence Review.

Chakravarthula, S. N., Gupta, R., Baucom, B. & Georgiou, P. (2015). A language-based generative model framework for behavioral analysis of couples' therapy.

Danilevsky, M., Qian, K., Aharonov, R., Katsis, Y., Kawas, B. & Sen, P. (2020). A Survey of the State of Explainable AI for Natural Language Processing.

Davis, F. (1989). Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. MIS Quarterly, vol. 13, pp. 319-340.

De Mauro, A., Greco, M. & Grimaldi, M. (2015). What is big data? A consensual definition and a review of key research topics. AIP Conference Proceedings 1644, 97. https://doi.org/10.1063/1.4907823

DeepAI. (n.d.). Hidden Layer [Machine Learning Glossary and Terms]. Retrieved April 8 2023 from: https://deepai.org/machine-learning-glossary-and-terms/hidden-layer-machine-learning

DeLone, W. H. & McLean, E. R. (1992). Information systems success: The quest for the dependent variable. MIS Quarterly, 16(2), 377-395.

Devlin, J., Chang, M.W., Lee, K. & Toutanova, K. (2018). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv preprint arXiv:1810.04805

Dillon, A. & Morris, M. (1996). User Acceptance of Information Technology: Theories and Models. Annual Review of Information Science and Technology. 31.

Egger, R. & Yu, J. (2022). A Topic Modeling Comparison Between LDA, NMF, Top2Vec, and BERTopic to Demystify Twitter Posts. Frontiers in sociology, 7, 886498. https://doi.org/10.3389/fsoc.2022.886498

Europol (2023). ChatGPT The impact of Large Language Models on Law Enforcement. Retrieved 5th March 2023 from: https://www.europol.europa.eu/cms/sites/default/files/documents/Tech%20Watch%20Flash%20-%20The%20Impact%20of%20Large%20Language%20Models%20on%20Law%20Enforcement.pdf

Fang, Z., Zhang, Q., Tang, X., Wang, A. & Baron, C. (2020). An implicit opinion analysis model based on feature-based implicit opinion patterns. Artificial Intelligence Review.

Feine, J., Morana, S. & Gnewuch, U. (2019). Measuring Service Encounter Satisfaction with Customer Service Chatbots using Sentiment Analysis. Wirtschaftsinformatik.

Fishbein, M. & Ajzen, I. (1975). Belief, attitude, intention and behaviour: An introduction to theory and research, Reading, Mass: Addison-Wesley Pub. Co, 1975.

Floridi, L. (2023). AI as Agency without Intelligence: on ChatGPT, large language models, and other generative models.

Forbes (2023). Microsoft Confirms Its $10 Billion Investment Into ChatGPT, Changing How Microsoft Competes With Google, Apple And Other Tech Giants. Retrieved 28 April 2023 from: https://www.forbes.com/sites/qai/2023/01/27/microsoft-confirms-its-10-billion-investment-into-chatgpt-changing-how-microsoft-competes-with-google-apple-and-other-tech-giants/

Fryer, L. K., Ainley, M., Thompson, A., Gibson, A. & Sherlock, Z. (2017). Stimulating and sustaining interest in a language course: An experimental. Computers in Human Behavior.

Gehring, J., Auli, M., Grangier, D., Yarats, D. & Dauphin, Y. N. (2017). Convolutional Sequence to Sequence Learning.

Goldstein, J. A., Sastry, G., Musser, M., DiResta, R., Gentzel, M. & Sedova, K. (2023). Generative Language Models and Automated Influence Operations: Emerging Threats and Potential Mitigations. arXiv preprint arXiv:2301.04246.

Gozalo-Brizuela, R. & Garrido-Merchan, E. C. (2023). ChatGPT is not all you need. A State of the Art Review of large Generative AI models. arXiv preprint arXiv:2301.04655.

Grover, P., Kar, A.K., Janssen, M. & Ilavarasan, P.V. (2019). Perceived usefulness, ease of use and user acceptance of blockchain technology for digital transactions – insights from user-generated content on Twitter, Enterprise Information Systems, 13:6, 771-800, DOI: 10.1080/17517575.2019.1599446

Grönroos, C. & Voima, P. (2013). Critical Service Logic: Making Sense of Value Creation and Co-Creation. Journal of the Academy of Marketing Science. 41. 133-150. 10.1007/s11747-012-0308-3.

Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F. & Pedreschi, D. (2018). A survey of methods for explaining black box.

Guo, B., Zhang, X., Wang, Z., Jiang, M., Nie, J., Ding, Y., Yue, J. & Wu, Y. (2023). How Close is ChatGPT to Human Experts? Comparison Corpus, Evaluation, and Detection. arXiv:2301.07597. Retrieved 9th of March 2023 from: https://www.arxiv-vanity.com/papers/2301.07597/

Hacker, P., Engel, A. & Mauer, M. (2023). Regulating chatgpt and other large generative ai models. arXiv preprint arXiv:2302.02337.

Hagger, M.S. (2019).The reasoned action approach and the theories of reasoned action and planned behavior. Oxford Bibliographies in Psychology, New York, Oxford University Press, 2019.

Haque, M. U., Dharmadasa, I., Sworna, Z. T., Rajapakse, R. N. & Ahmad, H. (2022). Exploring Sentiments of ChatGPT Early Adopters. arXiv:2212.05856

Helberger, N. & Diakopoulos, N. (2023). ChatGPT and the AI Act. Internet Policy Review, 12(1).

Hovy, D. & Prabhumoye, S. (2021). Five sources of bias in natural language. Language and Linguistics Compass, 15(8), e12432.

Hu, K. (2023). ChatGPT sets record for fastest-growing user base - analyst note. Reuters. Retrieved 28th March 2023 from: https://www.reuters.com/technology/chatgpt-sets-record-fastest-growing-user-base-analyst-note-2023-02-01/

Huang, K. (2023). Alarmed by A.I. Chatbots, Universities Start Revamping How They Teach. New York Times. Retrieved 24th April 2023 from: https://www.nytimes.com/2023/01/16/technology/chatgpt-artificial-intelligence-universities.html

Hunter, J.D. (2007). "Matplotlib: A 2D Graphics Environment", Computing in Science & Engineering, vol. 9, no. 3, pp.

Hutto, C.J. & Gilbert, E.E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. Eighth International Conference on Weblogs and Social Media (ICWSM-14). Ann Arbor, MI, June 2014.

Jordan, M.I. & Mitchell, T. (2015). Machine learning: Trends, perspectives, and prospects. Science, 349, 255 - 260. DOI: 10.1126/science.aaa8415

JustAnotherArchivist. (n.d.) snscrape. Retrieved 15th February 2023 from: https://github.com/JustAnotherArchivist/snscrape

Jurafsky, D. & Martin, J.H. (2000). Speech and language processing - an introduction to natural language processing, computational linguistics, and speech recognition. Prentice Hall series in artificial intelligence.

Kaity, M. & Balakrishnan, V. (2020). Sentiment lexicons and non-English languages: a survey. Knowledge and Information Systems.

Kalchbrenner, N., Lasse Espeholt, K. S., Oord, A. v. & Graves, A. (2017). Neural machine translation in linear time.

Khalil, M. & Er, E. (2023). Will ChatGPT get you caught? Rethinking of plagiarism detection. arXiv preprint arXiv:2302.04335.

Khan, Q. & Chua, H. N. (2021). Comparing Topic Modeling Techniques for Identifying Informative and Uninformative Content: A Case Study on COVID-19 Tweets. In 2021 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET) (pp. 1-6). Kota Kinabalu, Malaysia: IEEE. https://doi.org/10.1109/IICAIET51634.2021.9573878

Khurana, D., Koli, A., Khatter, K. & Singh, S. (2023). Natural language processing: state of the art, current trends and challanges. Multimedia Tools and Applications.

Kim, H.W., Kankanhalli, A. & Lee, H.L. (2016). Investigating decision factors in mobile application purchase: A mixed-methods approach," Information & Management. Vol. 53. No. 6. Pp. 727-739.

Kim, Y., Denton, C., Hoang, L. & Rush, A. M. (2017). Structured attention networks.

Kiss, I. M. & Buzás, N. (2015). Who Tweets About Technology?: Investigating the Role of Twitter in the Diffusion of Technological Information. International Journal of Knowledge and Systems Science (IJKSS), 6(1), 46-59.

Kocaballi, A. B., Quiroz, J. C., Rezazadegan, D., Berkovsky, S., Magrabi, F., Coiera, E. & Laranjo, L. (2020). Responses of conversational agents to health and lifestyle prompts: Investigation of appropriateness and presentation structures. Journal of Medical Internet Research.

Koçak, N. G., Kaya, S. & Erol, E. (2013). Social media from the perspective of diffusion of innovation approach. The Macrotheme Review, 2(3), 22-29.

Kohl, C., Mostafa, D., Böhm, M. & Helmut, K. (2017). Disruption of Individual Mobility Ahead? A Longitudinal Study of Risk and Benefit Perceptions of Self-Driving Cars on Twitter. Conference: Internationale Tagung Wirtschaftsinformatik (WI). St. Gallen, Switzerland.

Kozinets, R.V. (2002). The Field behind the Screen: Using Netnography for Marketing Research in Online Communities. Journal of Marketing Research, 39, 61 - 72.

Kung, T.H., Cheatham, M., Medenilla, A., Sillos, C., De Leon, L. & Elepaño, C. (2023). Performance of ChatGPT on USMLE: Potential for AI-assisted medical education using large language models. PLOS Digit Health 2(2): e0000198. doi.org/10.1371/journal.pdig.0000198

Kurzweil, R. (2004). The Law of Accelerating Returns. In: Teuscher, C. (eds) Alan Turing: Life and Legacy of a Great Thinker. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-05642-4_16

Kushwaha, A. K., Kumar, P. & Kar, A. K. (2021). What impacts customer experience for B2B enterprises on using AI-enabled. Industrial Marketing Management.

Kühl, N., Goutier, M., Baier, L., Wolff, C. & Martin, D. (2022). Human vs. supervised machine learning: Who learns patterns faster? Cognitive Systems Research, Volume 76, pp. 78-92, ISSN 1389-0417. https://doi.org/10.1016/j.cogsys.2022.09.002.

Lampo, A. (2023). How is Technology Accepted? Fundamental Works in User Technology Acceptance from Diffusion of Innovations to UTAUT-2. In Proceedings of the 8th International Conference on Industrial and Business Engineering (ICIBE '22). Association for Computing Machinery, New York, NY, USA, 260–266. https://doi-org.esc-web.lib.cbs.dk/10.1145/3568834.3568903

Law, M. (2022). From ELIZA to ChatGPT: The evolution of chatbots technology. Technology Magazine. Retrieved 12th March 2023 from: https://technologymagazine.com/articles/from-eliza-to-chatgpt-the-evolution-of-chatbots-technology

Leeja, M. & Bindu, V. (2020). A Review of Natural Language Processing Techniques for Sentiment Analysis using. Fourth International Conference on Computing Methodologies and Communication.

Leo Wong, D.Q. (2023). Singaporean Government Planning to Officially Teach Students How to Use ChatGPT. Retrieved 22nd April 2023 from: https://www.gizmochina.com/2023/02/22/singapore-students-chatgpt/

Li, H. (2018). Deep learning for natural language processing: advantages. Natl Sci Rev.

Li, L., Goethals, F.G., Giangreco, A. & Baesens, B. (2013). Using social network data to predict technology acceptance. International Conference on Interaction Sciences.

Lia, X., Q. X., Jiangb, J., Zhouc, Y. & Huang, L. (2019). Identifying and monitoring the development trends of emerging technologies using patent analysis and Twitter data mining: The case of perovskite solar cell technology. Technological Forecasting and Social Change.

Louho, R., Kallioja, M. & Oittinen, P. (2006). Factors affecting the use of Hybrid media applications. Graphic arts in Finland, 35 (3), pp. 11-21.

Luo, B., Lau, R. Y., Li, C. & Si, Y.-W. (2021). A critical review of state-of-the-art chatbot designs and applications; Yain-Whar Si. WIREs Data Mining and Knowledge Discovery.

Malhotra, C.K & Malhotra, A. (2015). MIT Sloan management review, 2016, Vol.57 (2), p.73.

Mandal, J.K., Mukherjee, I., Bakshi, S., Chatterji, S. & Sa, P.K. (2021). Computational Intelligence and Machine Learning. Advances in Intelligent Systems and Computing, vol 1276. Springer, Singapore. https://doi.org/10.1007/978-981-15-8610-1_7

McCallum, S. (2023). ChatGPT banned in Italy over privacy concerns. BBC. Retrieved 1st May 2023 from: https://www.bbc.com/news/technology-65139406

McGuffie, K. & Newhouse, A. (2020). The Radicalization Risks of GPT-3 and Advanced Neural Language Models.

Mehdi, Y. (2023). Reinventing search with a new AI-powered Microsoft Bing and Edge, your copilot for the web. Retrieved from 4th April 2023: https://blogs.microsoft.com/blog/2023/02/07/reinventing-search-with-a-new-ai-powered-microsoft-bing-and-edge-your-copilot-for-the-web/

Mhatre, M., Phondekar, D., Kadam, P., Chawath, A. & Ghag, K. (2017). Dimensionality reduction for sentiment analysis using pre-processing techniques. International

Conference on Computing Methodologies and Communication (ICCMC). Erode, India. Pp. 16-21. doi: 10.1109/ICCMC.2017.8282676.

Mitchell, R. (2018). Web scraping with Python: Collecting more data from the modern web. O'Reilly Media, Inc.

Ministry for Schools and Education in North Rhine-Westphalia (2023). Umgang mit textgenerierenden KI-Systemen. Retrieved 21st April 2023 from: https://www.schulministerium.nrw/system/files/media/document/file/handlungsleitfaden_ki_msb_nrw_230223.pdf

Mnif, E., Mouakhar, K. & Jarboui, A. (2021). Blockchain technology awareness on social media: Insights from twitter analytics. Journal of High Technology Management Research. https://doi.org/10.1016/j.hitech.2021.100416 R

Mohamad Suhaili, S., Salim, N. & Jambli, N.Z. (2021). Service chatbots: A systematic review. Expert Systems with Applications Vol. 184, 115461, ISSN 0957-4174. https://doi.org/10.1016/j.eswa.2021.115461.

Murphy, K. P. (2022). Probabilistic machine learning: an introduction. MIT press.

Neumann, M., Rauschenberger, M. & Schön, E. M. (2023). We Need To Talk About ChatGPT: The Future of AI and Higher Education.

Nikolenko, S. I., Koltcov, S. & Koltsova, O. (2015). Topic modelling for qualitative studies. Journal of Information Science, 43(1), 88-102.

Nilsson, N.J. (2009). The Quest for Artificial Intelligence. Cambridge University Press. DOI: 10.1017/cbo9780511819346

Nguyen, D. Q., Vu, T. & Nguyen, A. T. (2020). BERTweet: A pre-trained language model for English Tweets. Proceedings of the 2020 EMNLP (Systems Demonstrations).

O'Malley, J. (2022). Why is twitter so important? Reader's digest. Retrieved 18th April 2023 from: https://www.readersdigest.co.uk/lifestyle/technology/why-is-twitter-so-important

OpenAI. (2022). ChatGPT: Optimizing Language Models for Dialogue. Retrieved 10th April 2023 from: https://openai.com/blog/chatgpt/

OpenAI (2023a). GPT-4 deepens the conversation on Duolingo. Retrieved 16th April 2023 from: https://openai.com/customer-stories/duolingo

OpenAI (2023b). New AI classifier for indicating AI-written text. Retrieved 20th April 2023 from: https://openai.com/blog/new-ai-classifier-for-indicating-ai-written-text

OpenAI (2023c). GPT4. Retrieved 25th April 2023 from: https://openai.com/research/gpt-4

Oye, N. D., Iahad, N.A. & Ab.Rahim, N. (2012). The history of UTAUT model and its impact on ICT acceptance and usage by academicians. Educational Information Technology, DOI: 10.1007/s10639- 012-9189-9.

Pano, T. & Kashef, R. (2020). A complete VADER-based sentiment analysis of bitcoin (BTC) tweets during the era of COVID-19. Big Data and Cognitive Computing, 4(4), 33.

Pavlik, J. V. (2023). Collaborating With ChatGPT: Considering the implications of Generative Artificial Intelligence for Journalism and Media. Journalism & Mass Communication Educator.

Plisson, J., Lavrac, N. & Mladenic, D. (2004). A rule based approach to word lemmatization. In Proceedings of IS. Vol. 3. Pp. 83-86.

PowerBI (2023). Retrieved 2nd May 2023 from: https://powerbi.microsoft.com/en-au/

Przegalinska, A., Ciechanowski, L., Stroz, A., Gloor, P. & Mazurek, G. (2019). In bot we trust: A new methodology of chatbot performance measures. Business Horizons, 62(6), 785-797.

Radford, A., Narasimhan, K., Salimans, T. & Sutskever, I. (2018). Improving language understanding by generative pre-training.

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D. & Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners.

Radford, J. & Joseph, K. (2020). Theory in, theory out: the uses of social theory in machine learning for social science. Frontiers in big Data, 3, 18.

Raman, R., Veena, S. & Vachharajani, H. (2021). Adoption of online proctored examinations by university students during COVID19: Innovation diffusion study. Education and Information Technologies.

Rogers, E.M. (1962). Diffusion of Innovations, third ed. Free Press. New York.

Rogers, E.M. (2003). Diffusion of Innovations, fifth ed. Free Press. NewYork.

Roose, K. (2023). Don't Ban ChatGPT in Schools. Teach With It. New York Times. Retrieved 26th March 2023 from: https://www.nytimes.com/2023/01/12/technology/chatgpt-schools-teachers.html

Rosner, F., Hinneburg, A., Röder, M., Nettling, M. & Both, A. (2014). Evaluating topic coherence measures. arXiv preprint arXiv:1403.6397.

Saarijärvi, H., Kannan, P.K. & Kuusela, H. (2013). Value co-creation: theoretical approaches and practical implications. European Business Review. Vol. 25 No. 1. Pp. 6-19. https://doi.org/10.1108/09555341311287718

Saunders, M., Lewis, P. & Thornhill, A. (2009) Research Methods for Business Students. Pearson, New York.

Saunders, M., Lewis, P. & Thornhill, A. (2019). Research Methods for Business Students. Pearson Education Limited.

Shankland, S. (2023). Why We're Obsessed With the Mind-Blowing ChatGPT AI Chatbot. CNET. Retrieved 9th April 2023 from: https://www.cnet.com/tech/computing/why-were-all-obsessed-with-the-mind-blowing-chatgpt-ai-chatbot/

Shaw, A. (2018). Netnography: Ethical And Legal Concerns. Retrieved on 6th April 2023 from: https://www.strategic-planet.com/2018/01/netnography-ethical-legal-concerns/

Shen, Y., Heacock, L., Elias, J., Hentel, K. D., Reig, B., Shih, G. & Moy, L. (2023). ChatGPT and Other Large Language Models are Double Edge Swords.

Shuhidan, S.M. (2019). Extending Technology Acceptance Model to EPV Application. International Journal of Financial Research. Vol. 10, no. 5, pp. 398-410, 2019.

Sievert, S. (2018). Altair: Interactive statistical visualizations for python. Journal of Open Source Software, 3(32), 1057.

Sinha, N., Singh, P., Gupta, M. & Singh, P. (2020). Robotics at workplace: An integrated Twitter analytics – SEM based approach for behavioral intention to accept. International Journal of Information Management.

Sirk, C. (2020). Diffusion of Innovation: How Adoption of New Tech Spreads. Crm.org. Retrieved 15th January 2023 from: https://crm.org/articles/diffusion-of-innovations

Sobania, D., Briesch, M., Hanna, C. & Petke, J. (2023). An Analysis of the Automatic Bug Fixing Performance of ChatGPT. ArXiv, abs/2301.08653.

Sodré, W. d. & Duarte, J. C. (2023). Chatbot Optimization using Sentiment Analysis and Timeline Navigation. Revista de Informática Teórica e Aplicada.

Solaiman, I., Brundage, M., Clark, J., Askell, A., Herbert-Voss, A., Wu, J. & Wang, J. (2019). Release Strategies and the Social Impacts of Language Models. OpenAI.

SpaCy (2023). spaCy 101: Everything you need to know. Retrieved 29th April 2023 from: https://spacy.io/usage/spacy-101

Sparkes M., (2023). Google Bard advert shows new AI search tool making a factual error. Retrieved 1st of April 2023 from: https://www.newscientist.com/article/2358426-google-bard-advert-shows-new-ai-search-tool-making-a-factual-error/

Stackoverflow (2019). Evaluation of topic modeling: How to understand a coherence value / c_v of 0.4, is it good or bad?. Retrieved 4th April 2023 from: https://stackoverflow.com/questions/54762690/evaluation-of-topic-modeling-how-to-understand-a-coherence-value-c-v-of-0-4

Statista (2022). Number of Twitter users worldwide from 2019 to 2024. Retrieved 20th March 2023 from: https://www.statista.com/statistics/303681/twitter-users-worldwide/

Statista (2023a). Interest in generative AI on Google searches from February 2022 to February 2023 worldwide, by week. Retrieved 20th March 2023 from: https://www.statista.com/statistics/1367868/generative-ai-google-searches-worldwide/

Statista (2023b). Number of social media users worldwide from 2017 to 2027(in billions). Retrieved 20th March 2023 from: https://www.statista.com/statistics/278414/number-of-worldwide-social-network-users/

Stringer, A. & Wiggers, K. (2023). ChatGPT: Everything you need to know about the AI-powered chatbot. Retrieved 1st April 2023 from: https://techcrunch.com/2023/04/25/chatgpt-everything-you-need-to-know-about-the-ai-powered-chatbot/

Sudweeks, F. & Rafaeli, S. (1996). How do you get a hundred strangers to agree: Computer mediated communication and collaboration. Computer networking and scholarship in the 21st century university, 115-136.

Sun, J., Liao, Q.V., Muller, M., Agarwal, M., Houde, S., Talamadupula, K. & Weisz, J.D. (2022). Investigating Explainability of Generative AI for Code through Scenario-based Design. In 27th International Conference on Intelligent User Interfaces (IUI '22). Association for Computing Machinery, New York, NY, USA, 212–228. https://doi.org/10.1145/3490099.3511119

Sung, F., Yang, Y., Zhang, L., Xiang, T., Torr, P.H.S. & Hospedales, T. (2018). Learning to Compare: Relation Network for Few-Shot Learning. The Computer Vision Foundation. https://doi.org/10.48550/arXiv.1711.06025.

Suri, P. & Nihar R. (2017). Comparison between LDA & NMF for Event Detection from Large Text Stream Data. In Proceedings of the 2017 International Conference on Computational Intelligence and Communication Technology (pp. 246-251).

Susnjak, T. (2022). ChatGPT: The End of Online Exam Integrity?. arXiv preprint arXiv:2212.09292.

Taecharungroj, V. (2023). "What Can ChatGPT Do?" Analyzing Early Reactions to the. Big Data and Cognitive Computing.

The Economist (2023). The AI boom: lessons from history. How powerful new technologies transform economies. Retrieved 13th March 2023 from: https://www.economist.com/finance-and-economics/2023/02/02/the-ai-boom-lessons-from-history

Thelwall, M., Atkinson, P., Delamont, S., Cernat, A., Sakshaug, J. W. & Williams, R. A. (2020). Sentiment analysis. SAGE Publications Ltd.

Thorp, H. H. (2023). ChatGPT is fun, but not an author. Science, 379(6630). 313-313.

Tong, Z. & Zhang, H. (2016). A text mining research based on LDA topic modelling. In International conference on computer science, engineering and information technology. Pp. 201-210.

Trincieri, R. (2023). Google's Bard vs OpenAI's ChatGPT — Which of The Rival AI Chatbots is Better? Medium. Retrieved 7th April 2023 from: https://medium.com/ryan-trincieri-writes/googles-bard-vs-openai-s-chatgpt-which-of-the-rival-ai-chatbots-is-better-6f114cd96b38

Twitter (2023). Twitter API. Retrieved 3rd March 2023 from: https://developer.twitter.com/en/products/twitter-api

Vallance, C. (2023). AI could replace equivalent of 300 million jobs – report. BBC. Retrieved 14th April 2023 from: https://www.bbc.com/news/technology-65102150

Varanasi, L. (2023). ChatGPT could be a Stanford medical student, a lawyer, or a financial analyst. Here's a list of advanced exams the AI bot has passed so far. Business Insider. Retrieved 8th March 2023 from: https://www.businessinsider.com/list-here-are-the-exams-chatgpt-has-passed-so-far-2023-1?r=US&IR=T#stanford-medical-school-clinical-reasoning-final-6

Vasquez, J. (2018). Kylie Jenner Just Knocked $1.3 Billion Off Snapchat's Market Value. Time. Retrieved 9th January 2023 from: https://time.com/5170990/kylie-jenner-snapchat-stock-value/

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N. & Polosukhin, I. (2017). Attention is all you need. Advances in neural information processing systems, 30.

Venkatesh, V., Morris, M.G., Davis, G.B. & Davis, F.D. (2003). User Acceptance of Information Technology: Toward a Unified View. MIS Quarterly, vol. 27, no. 3, pp. 425-478.

Wallach, H. (2018). Computational social science≠ computer science+ social data. Communications of the ACM, 61(3), 42-44.

Wankhade, M., Rao, A. C. & Kulkarni, C. (2022). A survey on sentiment analysis methods, applications, and challenges. Artificial Intelligence Review.

Weisz, J.D., Muller, M., Houde, S., Richards, J., Ross, S.I., Martinez, F., Agarwal, M. & Talamadupula, K. (2021). Perfection Not Required? Human-AI Partnerships in Code Translation. In 26th International Conference on Intelligent User Interfaces. Pp. 402–412.

Weizenbaum, J. (1966). ELIZA—a computer program for the study of natural language communication between man and machine. Communications of the ACM, 9(1), 36-45.

White, J., Hays, S., Fu, Q., Spencer-Smith, J. & Schmidt, D.C. (2023). ChatGPT Prompt Patterns for Improving Code Quality, Refactoring, Requirements Elicitation, and Software Design. https://doi.org/10.48550/arXiv.2303.07839.

Widyaningrum, P., Ruldeviyani, Y. & Dharayani, R. (2019). Sentiment Analysis to Assess the Community's Enthusiasm Towards the Development Chatbot Using an Appraisal Theory. Procedia Computer Science.

Wirtz, B.W. & Göttel, V. (2016). Technology Acceptance in Social Media: Review, Synthesis and Directions for Future Empirical Research. Journal of Electronic Commerce Research, 17, 97.

Wohlin, C. (2014). Guidelines for snowballing in systematic literature studies and a replication in software engineering. In Proceedings of the 18th international conference on evaluation and assessment in software engineering (pp. 1-10)

Wood, P. (2023). Oxford and Cambridge ban ChatGPT over plagiarism fears but other universities choose to embrace AI bot. Inews. Retrieved 3rd March 2023 from: https://inews.co.uk/news/oxford-cambridge-ban-chatgpt-plagiarism-universities-2178391

Xu, Q. A., Chang, V. & Jayne, C. (2022). A systematic review of social media-based sentiment analysis: Emerging trends and challenges. Decision Analytics Journal, 2-4.

Yun, G., Cho, K., Jeong, Y. & Nam, T.-J. (2022). Ideasquares: Utilizing generative text as a source of design inspiration. DRS Biennial Conference Series.

Zahoor, S. & Rohilla, R. (2020). Twitter sentiment analysis using lexical or rule based approach: a case study. International Conference on Reliability, Infocom Technologies and Optimization (ICRITO) (Trends and Future Directions).

Zhai X. (2022). ChatGPT User Experience: Implications for Education. Available at SSRN: https://ssrn.com/abstract=4312418 or http://dx.doi.org/10.2139/ssrn.4312418

Zarifhonarvar, A. (2023). Economics of ChatGPT: A Labor Market View on the Occupational Impact of Artificial Intelligence. Indiana University. http://dx.doi.org/10.2139/ssrn.4350925

Zeng, X., Wang, F., Luo, Y., Kang, S.-g., Tang, J., Lightstone, F. C. & Fang, E. F. (2022). Deep generative molecular design reshapes drug discovery. Cell Reports Medicine.